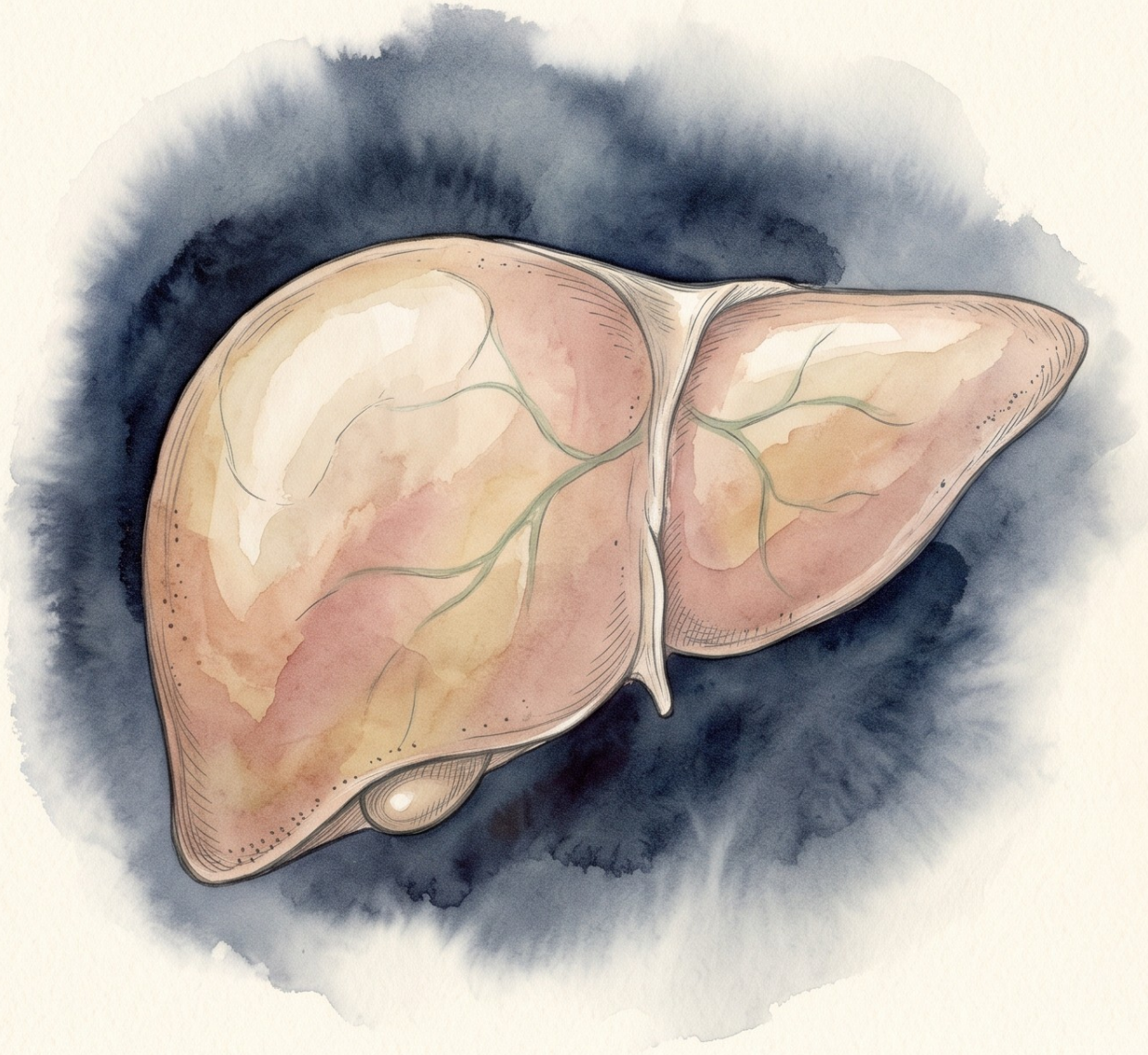


# Acute Liver Failure and ACLF in the ICU

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Assistant Professor of Medicine and Surgery
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Tufts University | School of Medicine

Slides (PDF)

*Illustrations generated with Google Gemini*

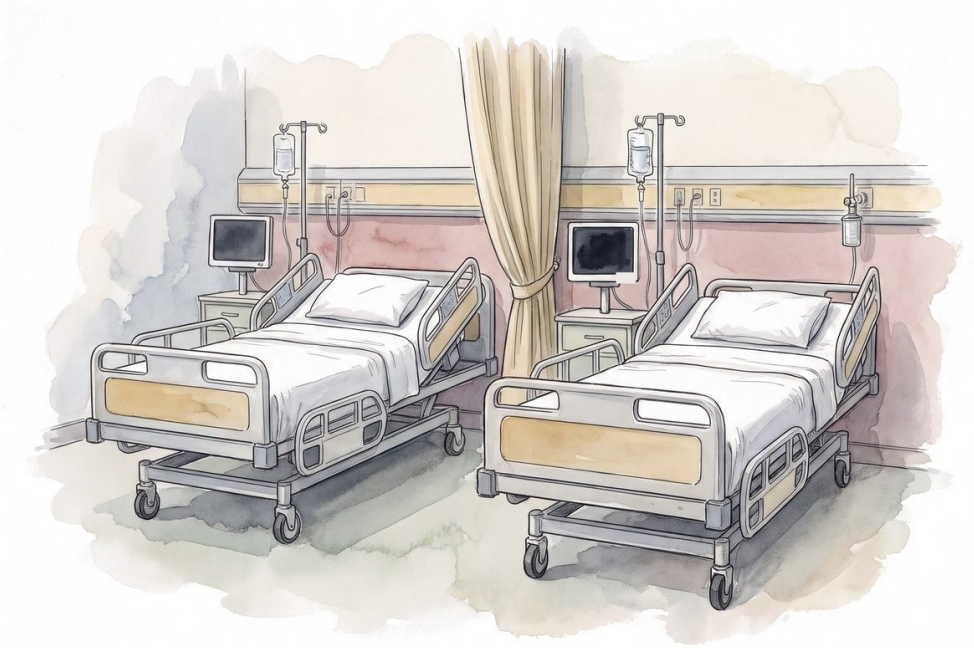
# Objectives

*By the end of this talk you should be able to:*

- Tell acute liver failure apart from severe acute liver injury and from ACLF
- Plan the first hour of care in ALF
- Manage the brain and the ammonia in ALF
- Recognize and grade ACLF, and find the trigger
- Work through AKI in cirrhosis
- Know when to call transplant hepatology

# Bed 4 and bed 7

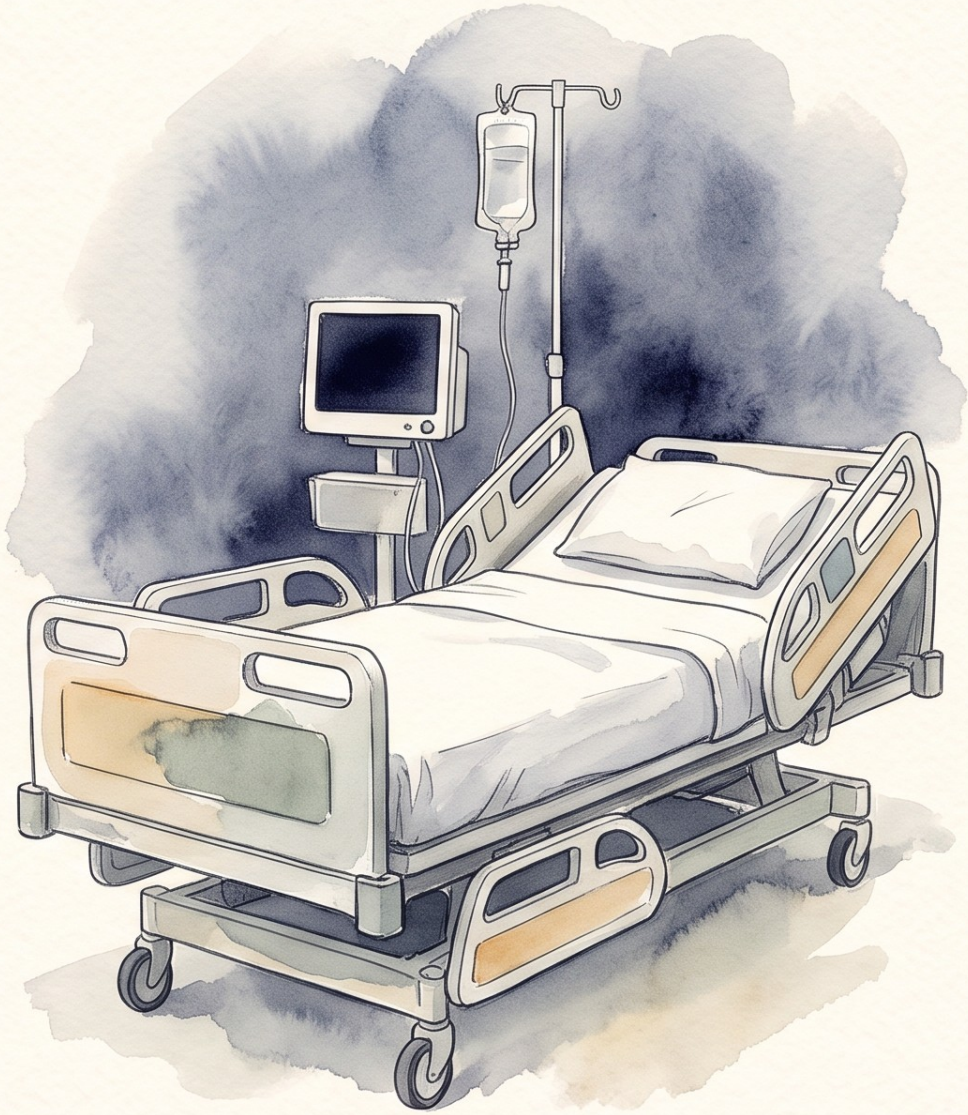
| Bed 4 | Bed 7 |
|---|---|
| 28-year-old woman | 56-year-old man |
| No known liver disease | Alcohol-related cirrhosis with ascites |
| Nausea and vomiting for 4 days | Fever, confusion, low blood pressure |



*Is it the same disease?*

# Bed 4

28F, 4 days of nausea



# 28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| ALP | 128 U/L |
| Bilirubin | 2.9 mg/dL |
| Creatinine | 1.0 mg/dL |

**Is this acute liver failure?**

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**Is this acute liver failure?**

**We don't know yet. What's missing?**

# 28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

| Test | Result |
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| AST | 6,820 U/L |
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| Creatinine | 1.0 mg/dL |

## What's missing?

- **INR** (synthetic function, not enzymes)
- **Mental status** (any encephalopathy?)
- **Time course** (how long since the first symptom?)
- **Is there chronic liver disease underneath?**

# 28F, 4 days of nausea

- Disoriented to time, asterixis. Still there after the glucose is fixed
- Symptoms started 4 days ago
- Ultrasound: smooth liver, no ascites, patent vessels, normal spleen
- No history or imaging suggesting cirrhosis

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| Bilirubin | 2.9 mg/dL |
| **INR** | **3.2** |
| **Glucose** | **54 mg/dL** |

# 28F, 4 days of nausea

- Disoriented to time, asterixis. Still there after the glucose is fixed
- Symptoms started 4 days ago
- Ultrasound: smooth liver, no ascites, patent vessels, normal spleen
- No history or imaging suggesting cirrhosis

| Test | Result |
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| ALT | 5,940 U/L |
| Bilirubin | 2.9 mg/dL |
| **INR** | **3.2** |
| **Glucose** | **54 mg/dL** |

**This is acute liver failure**

# Acute liver failure

- **INR ≥ 1.5** and **any degree of encephalopathy**
- Illness of less than 26 weeks
- No pre-existing cirrhosis
  - AASLD and ACG: chronic liver disease without cirrhosis still allows ALF. EASL: no prior liver disease at all
  - Wilson disease, autoimmune hepatitis and vertically acquired HBV can still count if recognized within 26 weeks

| Jaundice to encephalopathy | Hyperacute (< 1 week) | Acute (1 to 4 weeks) | Subacute (4 to 12 weeks) |
|---|---|---|---|
| Typical causes | Acetaminophen, HAV, HEV, ischemia | HBV | DILI, AIH, unknown cause |
| Cerebral edema | **Common** | Less common | Uncommon |
| Spontaneous recovery | More likely | Intermediate | **Less likely** |

*EASL ALF guideline 2017 [PMID: 28417882]; ACG ALF 2023 [PMID: 37377263]; Management of ALF: update 2022 [PMID: 36001996]*

# Same patient, but wide awake

*Same labs and INR 2.4, but fully oriented with no asterixis*

**What do you call this, and what do you do now?**



# Same patient, but wide awake

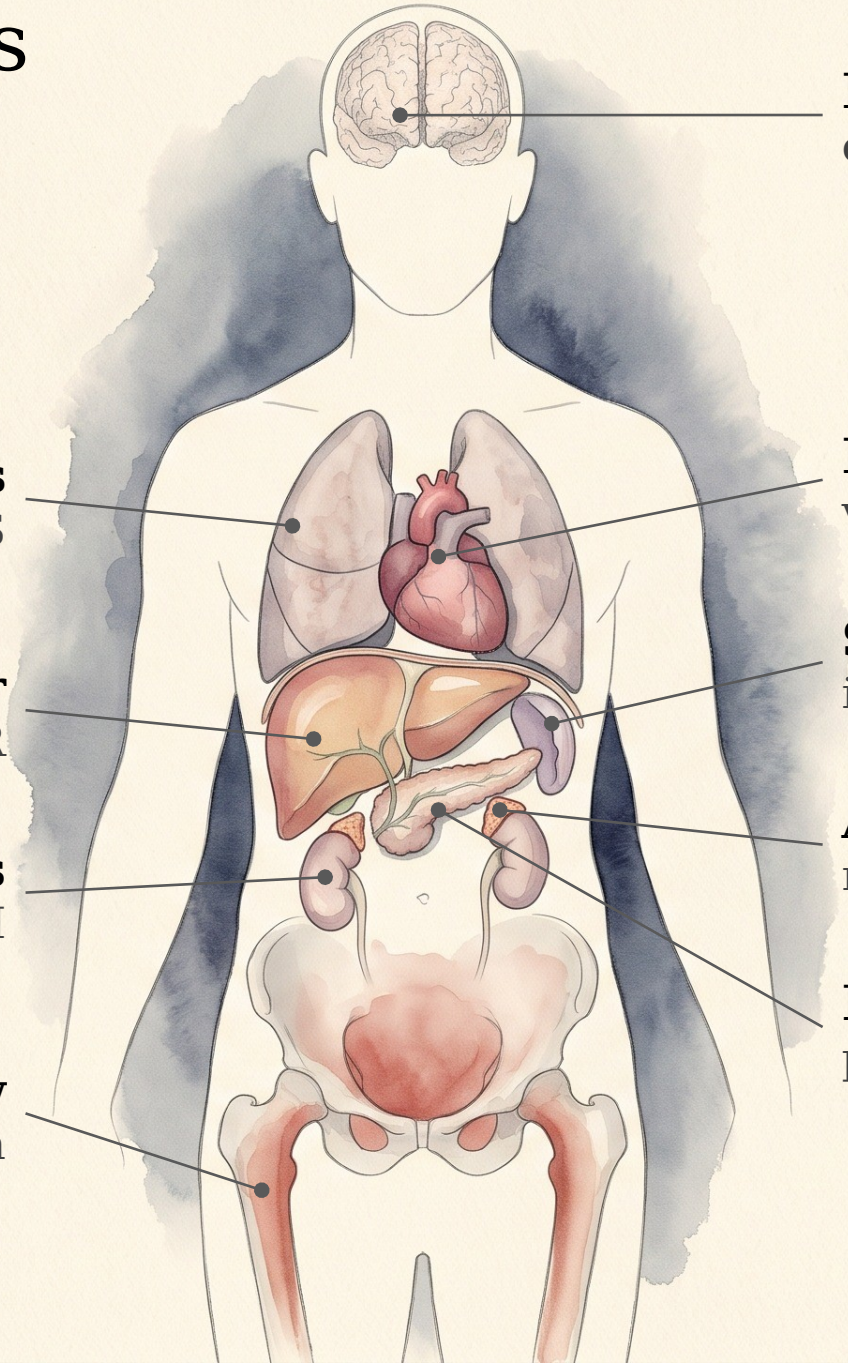
*Same labs and INR 2.4, but fully oriented with no asterixis*

- **Severe acute liver injury**, not ALF yet
- No encephalopathy, so it doesn't meet the definition
- But it can become ALF within hours
  - Check glucose often and examine the patient every few hours
- And the opposite trap:
  - Hypoglycemia, sedatives, sepsis, alcohol withdrawal and seizures can all look like encephalopathy

# Which other organs fail in ALF?



# Which other organs fail in ALF?



**Brain**
encephalopathy, cerebral edema

**Lungs**
ARDS

**Heart and vessels**
vasodilatory shock

**Liver**
hypoglycemia, lactate, INR

**Spleen and immune system**
infection

**Kidneys**
AKI

**Adrenals**
relative insufficiency

**Pancreas**
pancreatitis

**Bone marrow**
suppression

*Figure content after Bernal W, Wendon J. N Engl J Med 2013;369:2525-34*

28F, 4 days of nausea

**What's your next step?**

# The first hour

| Support | Workup |
|---|---|
| ICU bed | Acetaminophen level |
| Glucose every 1 to 2 hours | Viral, autoimmune, Wilson, toxicology |
| Volume, perfusion, lactate | Doppler ultrasound |
| Neuro checks | Pregnancy test |

**Send the workup while you resuscitate**

# Which cause changes treatment right away?

| If you suspect | Clues | Send | Don't wait to |
|---|---|---|---|
| Acetaminophen | AST/ALT in the thousands, OTC use | Level, combination products | |
| Herpes simplex | Fever, very high AST/ALT, little jaundice | HSV PCR (blood) | |
| Hepatitis B | Endemic area, chemo or immunosuppression | HBsAg, anti-HBc IgM, DNA | |
| Autoimmune hepatitis | Woman, other autoimmune disease | ANA, ASMA, IgG | |
| Wilson disease | Under 40, Coombs-negative hemolysis | Ceruloplasmin, urine copper | |
| AFLP, HELLP | Third trimester, high BP, low platelets | hCG | |
| Budd-Chiari | Pain, ascites, big liver | Doppler or CT | |
| Ischemic hepatitis | Shock or arrest, very high LDH | Echo | |
| Amanita | Wild mushrooms, diarrhea 6-24 h later | Poison control | |

*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]; EASL AIH 2025 [PMID: 40348684]*

# Which cause changes treatment right away?

| If you suspect | Clues | Send | Don't wait to |
|---|---|---|---|
| Acetaminophen | AST/ALT in the thousands, OTC use | Level, combination products | Start NAC |
| Herpes simplex | Fever, very high AST/ALT, little jaundice | HSV PCR (blood) | Start IV acyclovir |
| Hepatitis B | Endemic area, chemo or immunosuppression | HBsAg, anti-HBc IgM, DNA | Entecavir or tenofovir |
| Autoimmune hepatitis | Woman, other autoimmune disease | ANA, ASMA, IgG | Steroids with hepatology |
| Wilson disease | Under 40, Coombs-negative hemolysis | Ceruloplasmin, urine copper | Urgent transplant |
| AFLP, HELLP | Third trimester, high BP, low platelets | hCG | Obstetrics: delivery |
| Budd-Chiari | Pain, ascites, big liver | Doppler or CT | Anticoagulate; TIPS |
| Ischemic hepatitis | Shock or arrest, very high LDH | Echo | Fix the circulation |
| Amanita | Wild mushrooms, diarrhea 6-24 h later | Poison control | NAC, silibinin |

*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]; EASL AIH 2025 [PMID: 40348684]*

# 28F, 4 days of nausea

- Roommate: "She was taking a lot of the Tylenol with codeine, plus cold medicine, for 3 days"
- Acetaminophen level: **12 µg/mL**, time of last dose unknown

**Can you use the nomogram?**



# 28F, 4 days of nausea

- Roommate: "She was taking a lot of the Tylenol with codeine, plus cold medicine, for 3 days"
- Acetaminophen level: **12 µg/mL**, time of last dose unknown



## Can you use the nomogram?

- **No.** Rumack-Matthew is for a single acute ingestion with a known time
- Repeated supratherapeutic use and late presentation: the level can be low or undetectable
- **Start NAC now**

# NAC for every kind of acute liver failure?

Published in final edited form as:
*Gastroenterology*. 2009 September ; 137(3): 856–864.e1. doi:10.1053/j.gastro.2009.06.006.

**WM Lee**[1], **LS Hynan**[1], **L Rossaro**[2], **RJ Fontana**[3], **RT Stravitz**[4], **AM Larson**[5], **TL Davern II**[6], **NG Murray**[7], **T McCashland**[8], **JS Reisch**[1], **PR Robuck**[9], and **the Acute Liver Failure Study Group**

**What do you think they found?**

# NAC for every kind of acute liver failure?

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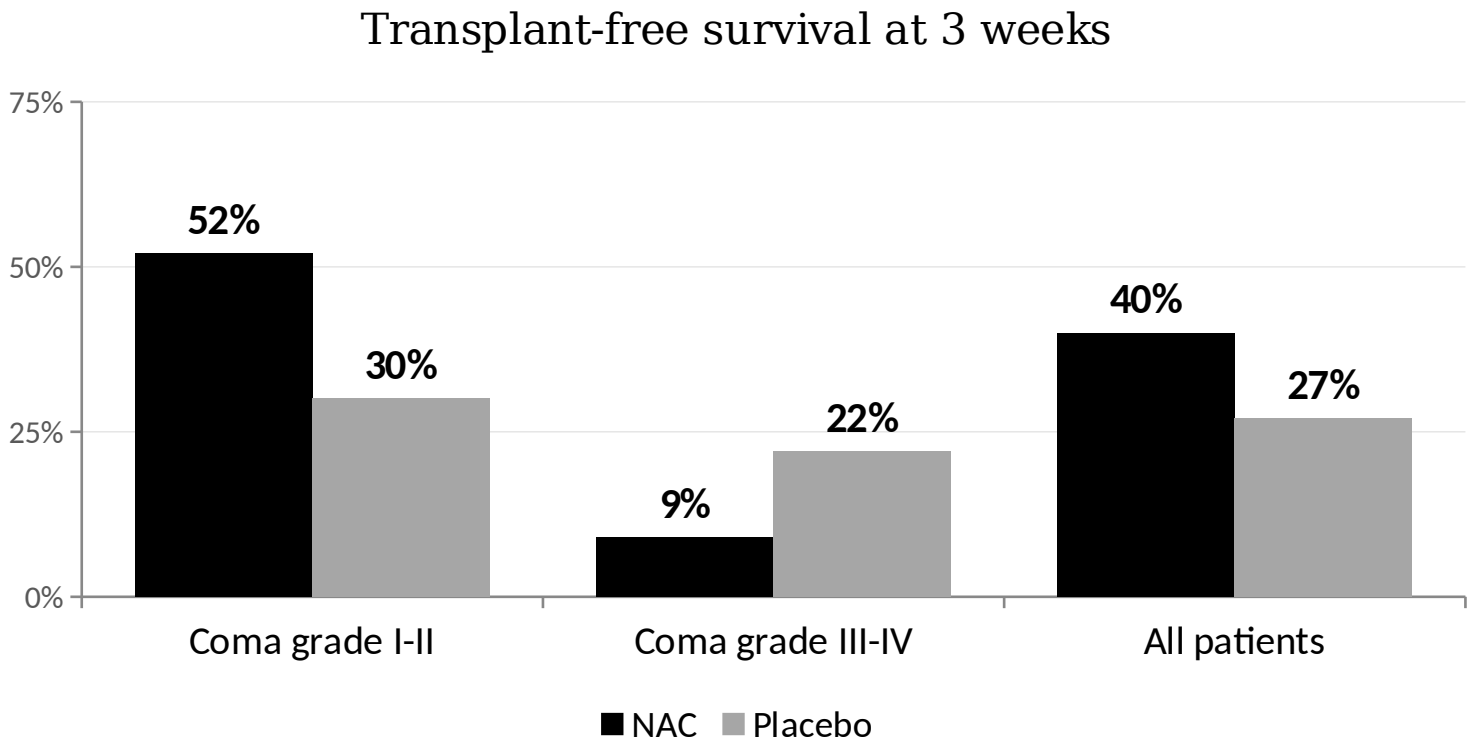
## INTRAVENOUS *N*-ACETYLCYSTEINE IMPROVES TRANSPLANT-FREE SURVIVAL IN EARLY STAGE NON-ACETAMINOPHEN ACUTE LIVER FAILURE

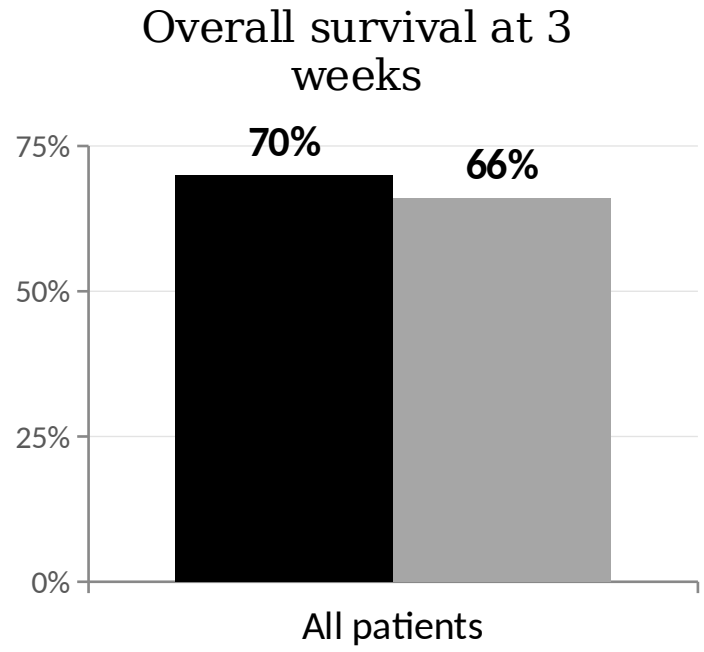
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**Conclusions—**Intravenous NAC improves transplant-free survival in patients with early stage non-acetaminophen related acute liver failure. Patients with advanced coma grades do not benefit from NAC and typically require emergency liver transplantation. (ClinicalTrials.gov number NCT00004467)

# NAC for every kind of acute liver failure?

*Adult non-acetaminophen ALF, randomized, NAC vs placebo, 173 patients*




**Reasonable in early non-acetaminophen ALF. Don't extrapolate to advanced coma, children or ACLF**

*Lee WM et al. Gastroenterology 2009 [PMID: 19524577]*

# 28F, 4 days of nausea

| | Admission | Hour 12 |
|---|---|---|
| Mental status | Disoriented, asterixis | **Responds only to voice (grade III)** |
| Ammonia (arterial) | Not sent | **182 µmol/L** |
| Urine output | OK | 15 mL/h for 4 hours |
| Lactate (after 3 L) | 2.1 mmol/L | 4.6 mmol/L |
| Creatinine | 1.0 mg/dL | 1.3 mg/dL |
| Sodium | 136 mmol/L | 133 mmol/L |
| Glucose | 54 mg/dL | Needs D10 |

## What worries you most?

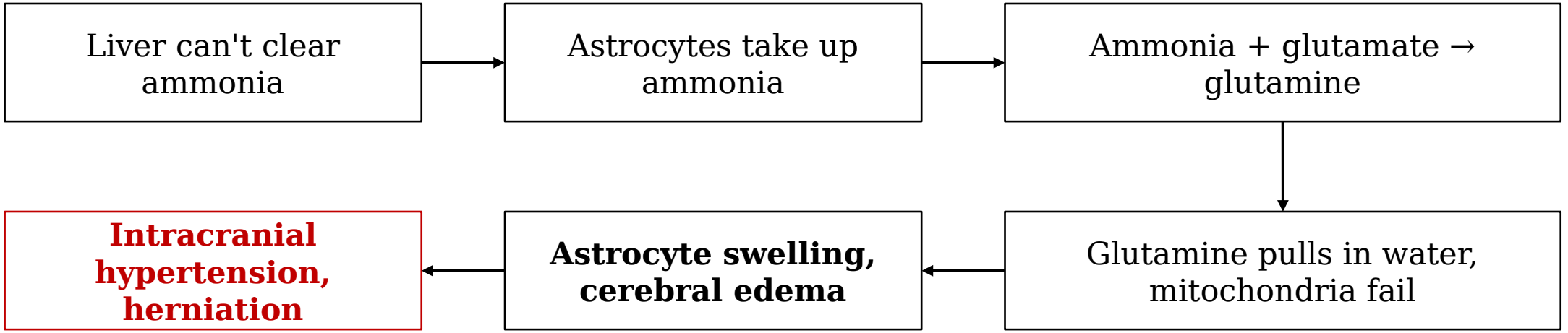
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## Her brain

*When do you intubate? Do you wait for the creatinine before dialysis?*

# Why does the brain swell?




- **Makes it worse:** inflammation and infection, loss of autoregulation, fever, low sodium
- **Highest risk:** hyperacute course, young, grade III-IV encephalopathy, ammonia persistently above 150 to 200 µmol/L, AKI, vasopressors

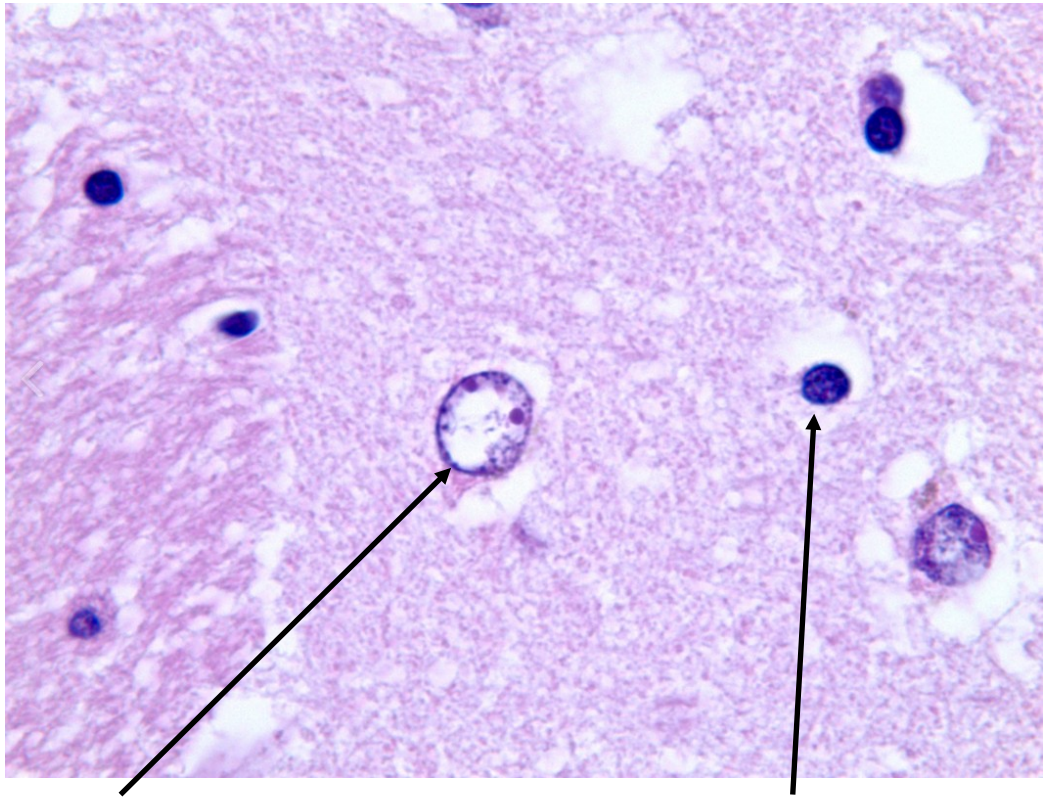
*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]*

# Ammonia: ALF vs cirrhosis



**Why do patients with cirrhosis rarely get cerebral edema, even with a high ammonia?**

# Ammonia: ALF vs cirrhosis



**Alzheimer type II astrocyte:** big, pale nucleus

**Normal glial nucleus:** small and dark

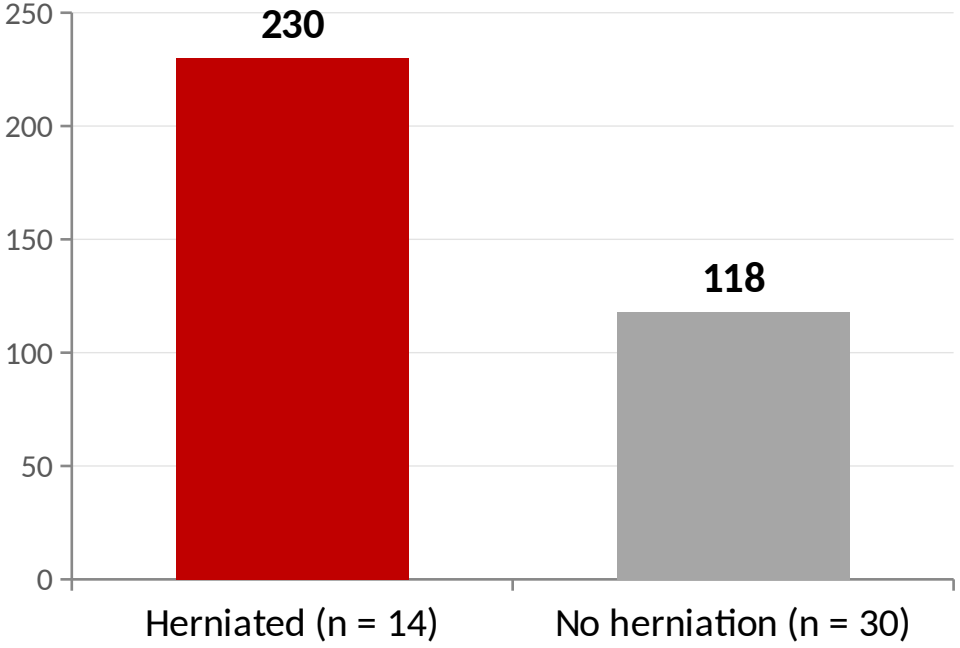
- Astrocytes clear ammonia by turning it and glutamate into glutamine
- Glutamine pulls water into the cell
- **Cirrhosis:** over months the astrocyte pushes out other solutes (myo-inositol, taurine) to make room. It swells only a little. This is the cell you see under the microscope
- **ALF:** the ammonia comes in days. No time to adapt, so the astrocytes swell and the brain swells with them

**Cerebral edema is common in hyperacute ALF and rare in cirrhosis**

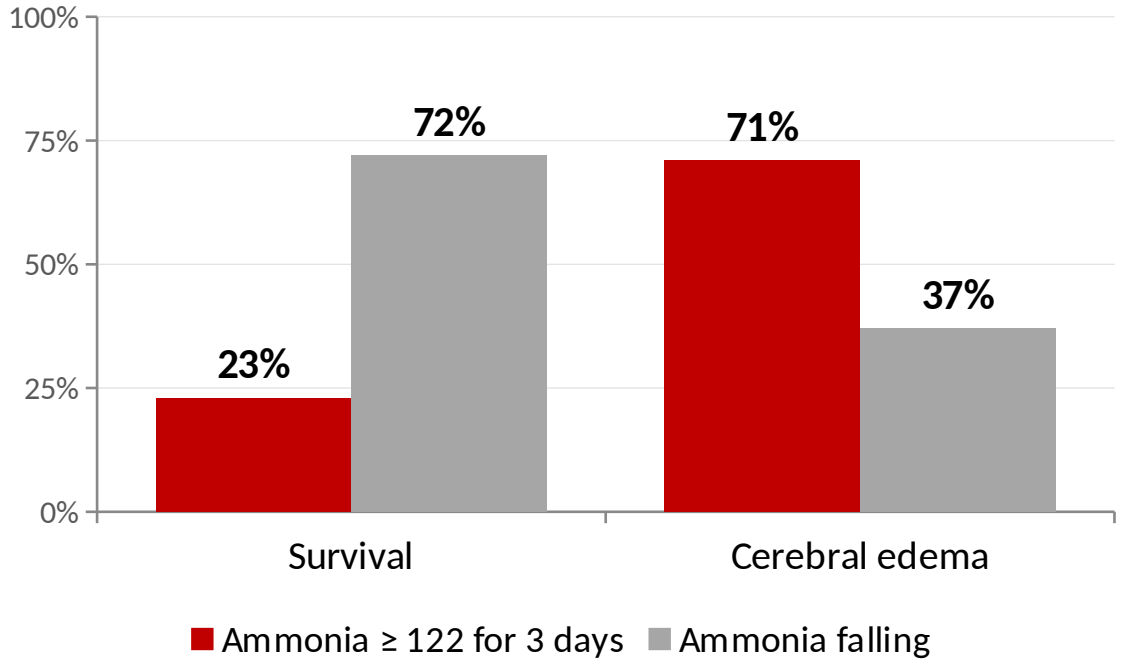



# Ammonia: how high is too high?

Clemmesen 1999: ammonia, µmol/L




Kumar 2012: 295 patients



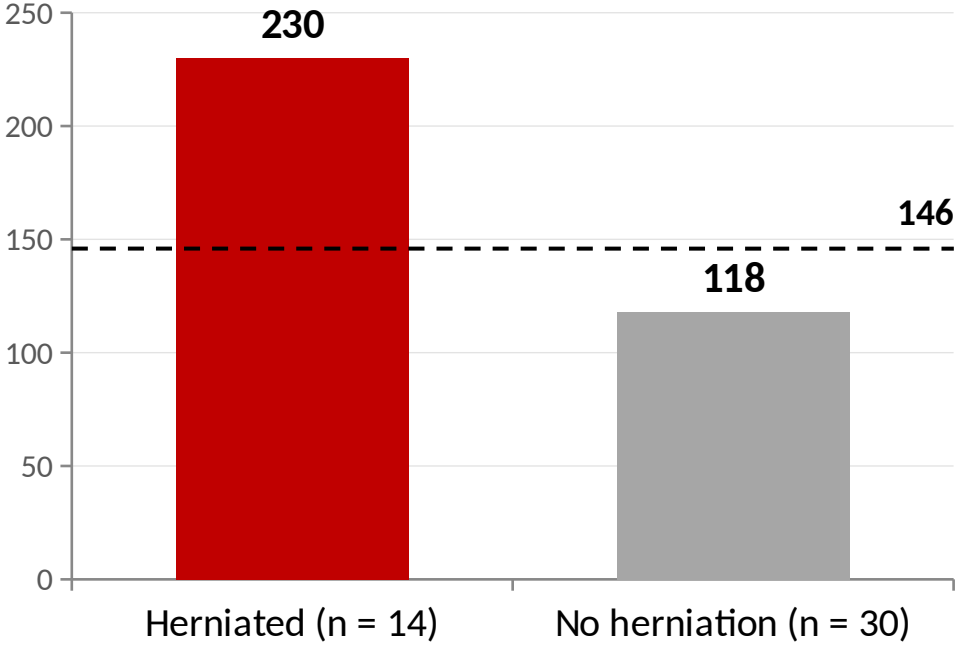

**So what number worries you?**

*Clemmesen JO et al. Hepatology 1999 [PMID: 10051463]; Kumar R et al. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]; more studies in the appendix*

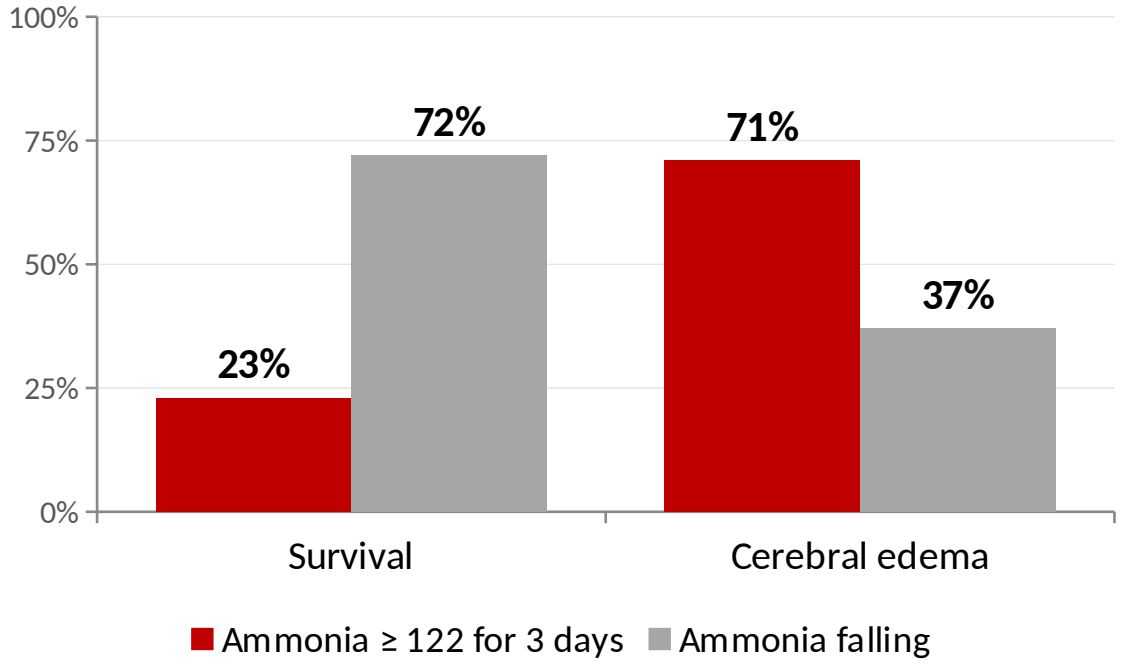



# Ammonia: how high is too high?

Clemmesen 1999: ammonia, µmol/L




Kumar 2012: 295 patients




**Above 150 to 200 and not falling: worry about the brain**

*No single level rules ICH in or out. Read it with the HE grade and the trend*

*Clemmesen JO et al. Hepatology 1999 [PMID: 10051463]; Kumar R et al. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]; more studies in the appendix*

# Lactulose for her?

*Hour 12: ammonia 182 µmol/L, grade III. In a cirrhotic we'd already be giving it*

**Yes or no?**



# Lactulose for her?

*Hour 12: ammonia 182 µmol/L, grade III. In a cirrhotic we'd already be giving it*

| Why people give it | Why not |
|---|---|
| Ammonia drives brain swelling | **No proven benefit in ALF (ALFSG)** |
| It's what we do in cirrhosis | Distended bowel gets in the way at transplant |
| Cheap; AASLD 2011 allowed it early | Diarrhea: free water loss, sodium swings |
| | Liquid by mouth or tube: aspiration risk |

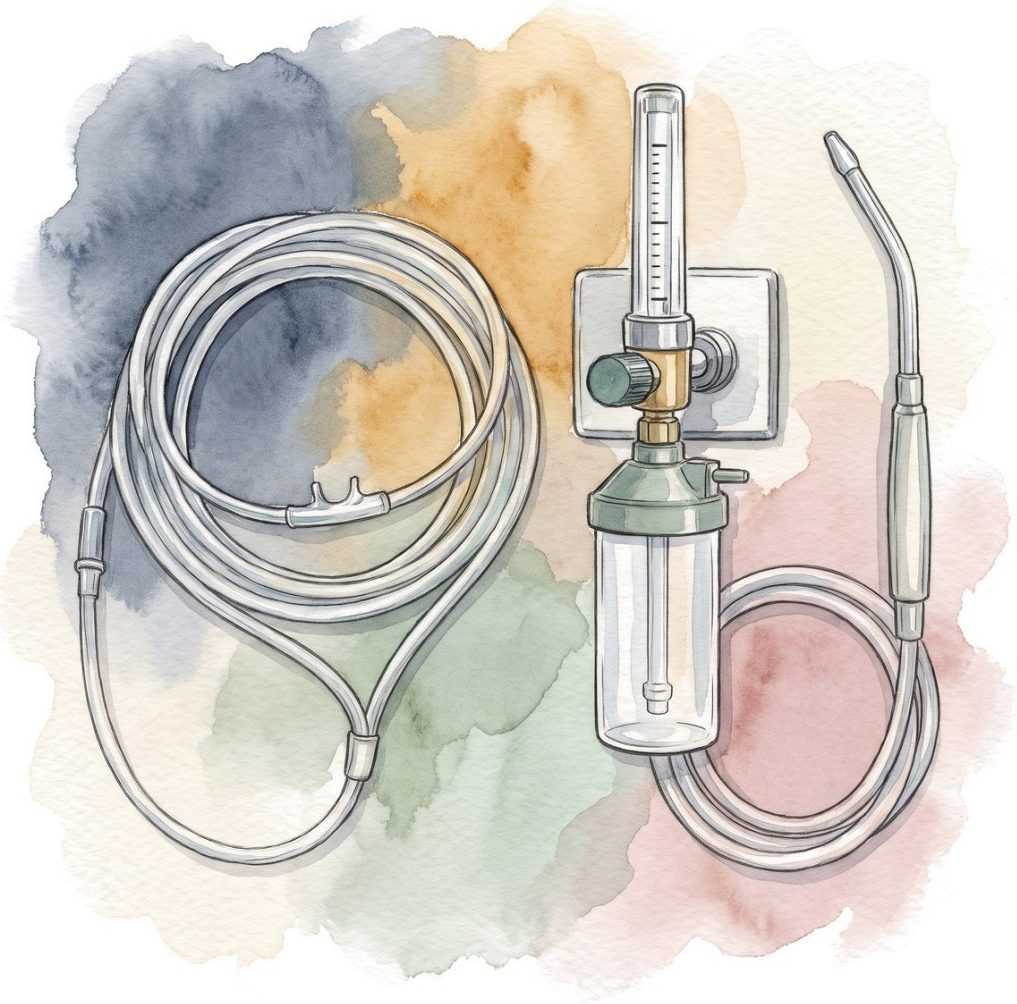
**Not for her**

*AASLD position paper 2011, rec. 26; Alba L, Hay JE, Lee WM et al. J Hepatol 2002;36(Suppl 1):33 (abstract); ACG ALF 2023 [PMID: 37377263]*

# When do you intubate?

- Responds to voice, drifts off, intermittently agitated
- Handling secretions for now
- SpO2 96% on 2 L

**Intubate now, or wait until she can't protect her airway?**



# When do you intubate?

## Grade III-IV encephalopathy in ALF: intubate

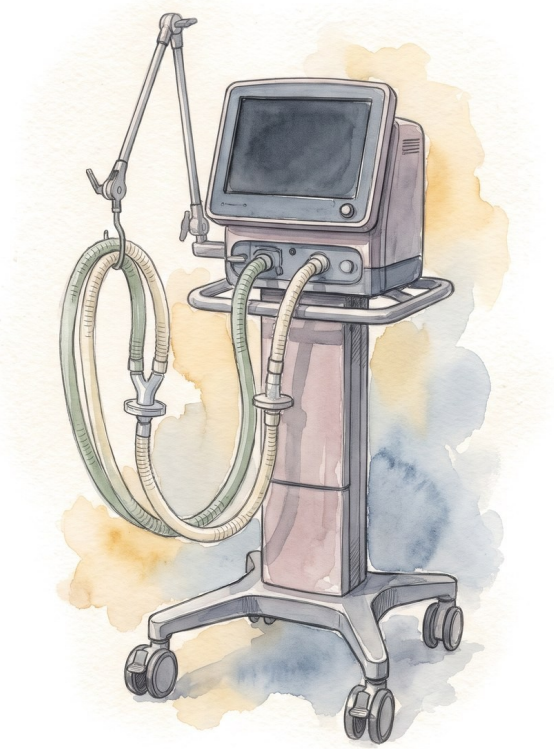
| Before | During | After |
|---|---|---|
| Preoxygenate<br>Norepinephrine ready<br>Check glucose<br>Head of bed 30°, neck neutral | Avoid hypotension (CPP)<br>No coughing or bucking (ICP)<br>Most experienced operator | Propofol: short-acting, may lower ICP<br>Avoid long-acting benzodiazepines<br>Check pupils: sedation hides the exam |

*Practical update 2024 [PMID: 39170946]; ACG ALF 2023 [PMID: 37377263]; SCCM 2020 [PMID: 32058387]*

# Can we use permissive hypercapnia?

*After intubation: bilateral infiltrates, P/F 150 on PEEP 10*

**Lung protection or brain protection?**



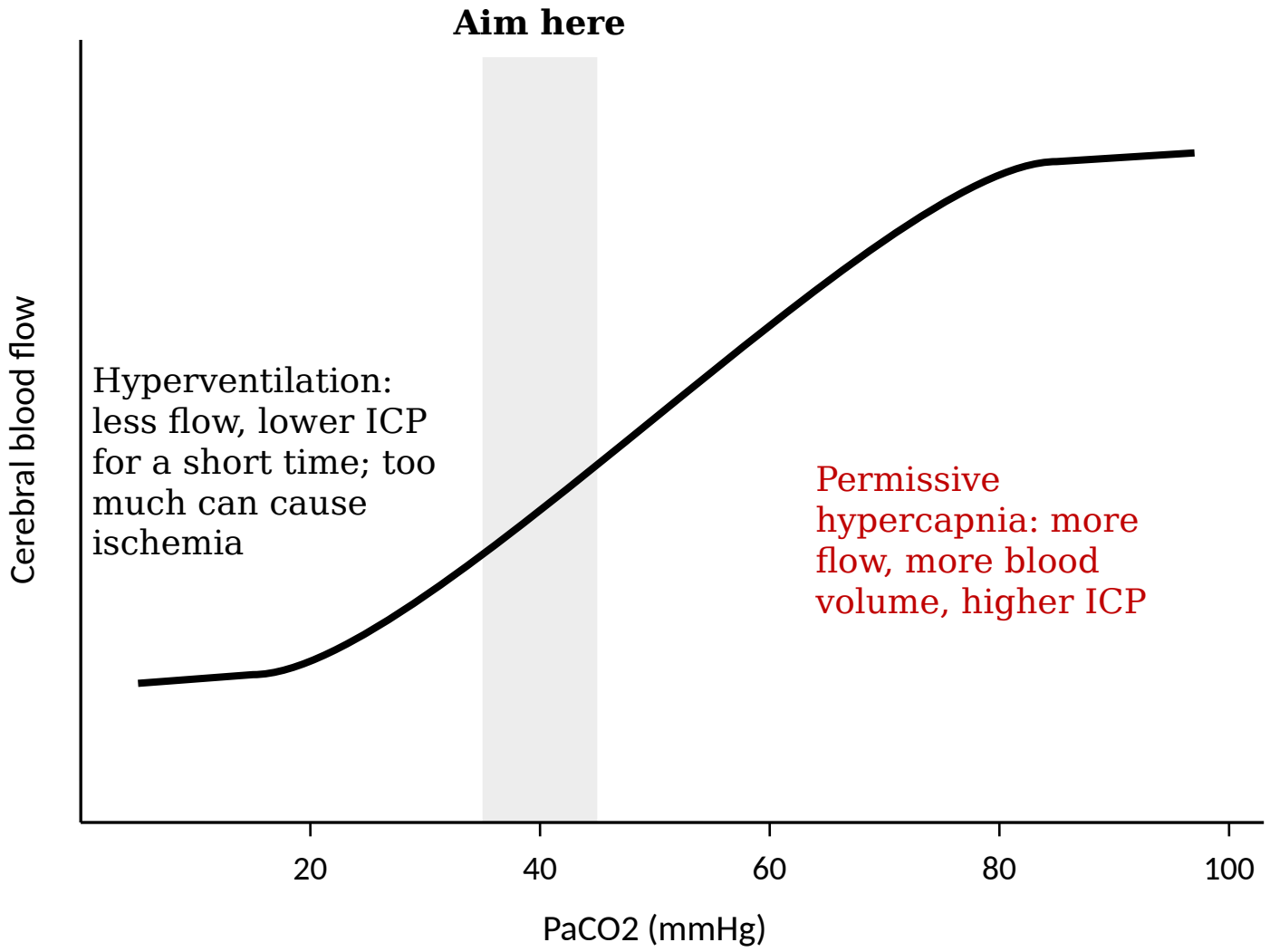
# Can we use permissive hypercapnia?

*After intubation: bilateral infiltrates, P/F 150 on PEEP 10*

| | For the lungs (ARDS) | For the brain (ICP) | What we do |
|---|---|---|---|
| **Tidal volume** | 6 mL/kg PBW | Watch the CO2 | 6 mL/kg, raise the rate |
| **CO2** | Tolerate a high CO2 | High CO2 raises ICP | **Keep the CO2 normal** |
| **Hyperventilation** | No | Lowers ICP briefly | Short rescue for herniation only |
| **PEEP** | Recruit | High PEEP: less venous return | Set for O2 and BP; PEEP is fine |

*SCCM 2020 [PMID: 32058387]; practical update 2024 [PMID: 39170946]*

# CO2 and cerebral blood flow

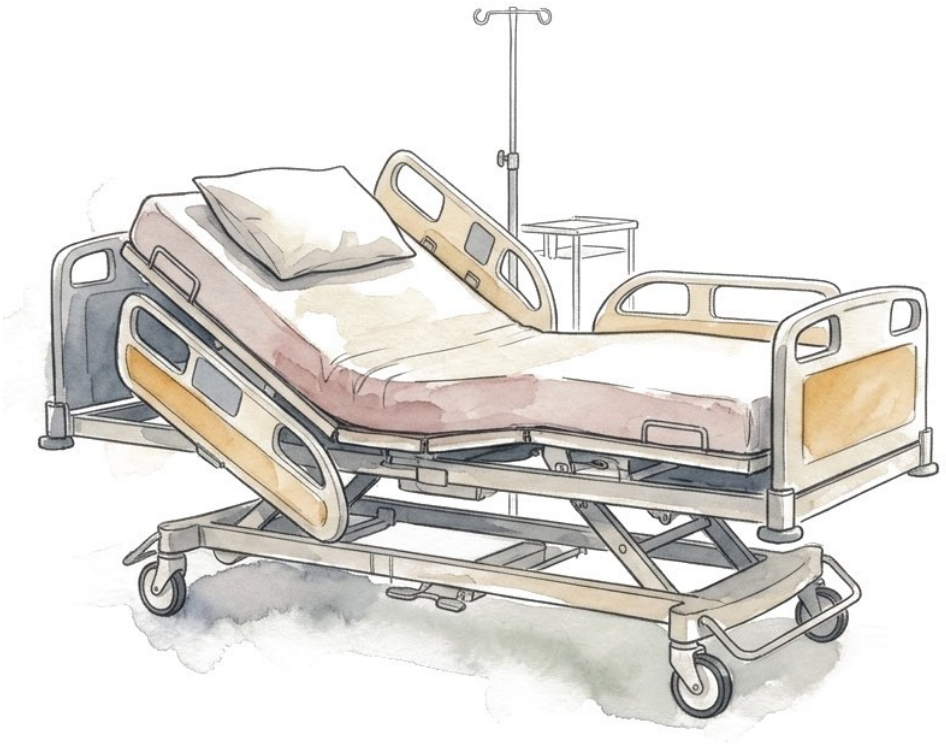



- Between about 20 and 80 mmHg, cerebral blood flow rises with PaCO2
- In a swollen brain, more blood volume means a higher ICP
- Keep PaCO2 normal. Hyperventilate only as a short bridge for herniation

*Schematic, not to scale. Standard cerebral physiology*

# Protecting the brain: every patient with grade III-IV

- Head of bed 30°, neck neutral
- Treat fever, avoid hypoglycemia
- Sodium **145 to 155 mmol/L** with hypertonic saline (use your center's protocol)
- Avoid fluid overload
- Treat seizures, EEG if in doubt. No routine prophylaxis
- No cooling: 34 °C did not prevent intracranial hypertension in an RCT



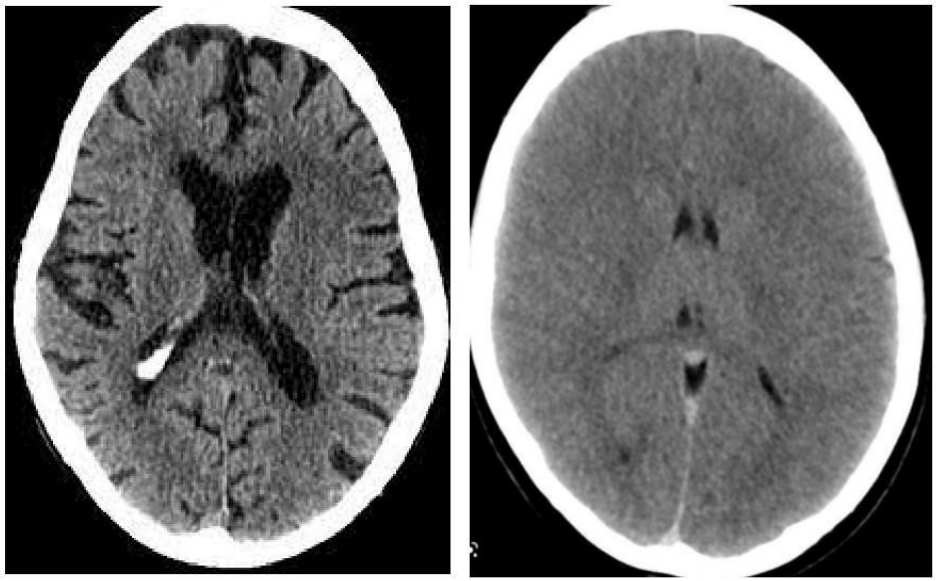
*SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]; Bernal W et al. J Hepatol 2016;65:273-9*

# Rising ICP: hypertension, bradycardia, a new pupil

- Hyperosmolar bolus (hypertonic saline or mannitol)
- Brief hyperventilation as a bridge
- Call the transplant team now

**Things to know**

- Hypertension and bradycardia come late. Don't wait for them
- A normal head CT does not exclude raised ICP
- ICP monitors are not routine; SCCM suggests against (very low certainty)



Normal | Effacement of sulci

*SCCM 2023 [PMID: 37052436]; practical update 2024 [PMID: 39170946]*

# The creatinine is only 1.3

*Ammonia 182 → 214 µmol/L, grade III, urine output 15 mL/h*

**Do you wait for a conventional dialysis indication?**

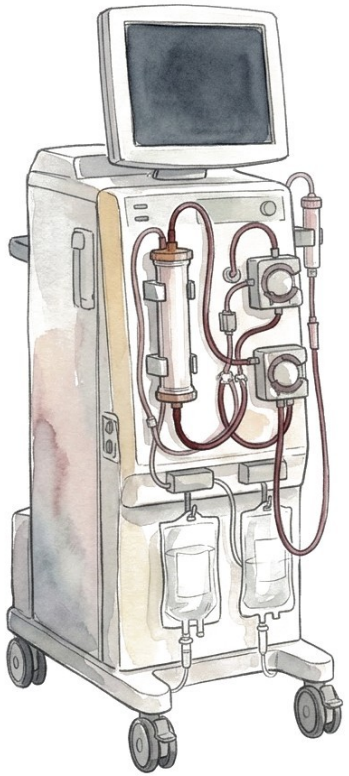


# The creatinine is only 1.3

*Ammonia 182 → 214 µmol/L, grade III, urine output 15 mL/h*

**No. In ALF, start CRRT for the ammonia**

- ACG 2023: suggests early CRRT for hyperammonemia with grade II or higher encephalopathy, even without a renal indication (conditional, very low certainty)
- Continuous, not intermittent: fewer swings in blood pressure and brain water
- Call nephrology now



*ACG ALF guideline 2023 [PMID: 37377263]; update 2022 [PMID: 36001996]*

# Should we correct this INR?

*Hour 12: INR 4.1, platelets 88,000. She needs a dialysis catheter*

## 2 units of FFP first?



# Should we correct this INR?

*Hour 12: INR 4.1, platelets 88,000. She needs a dialysis catheter*

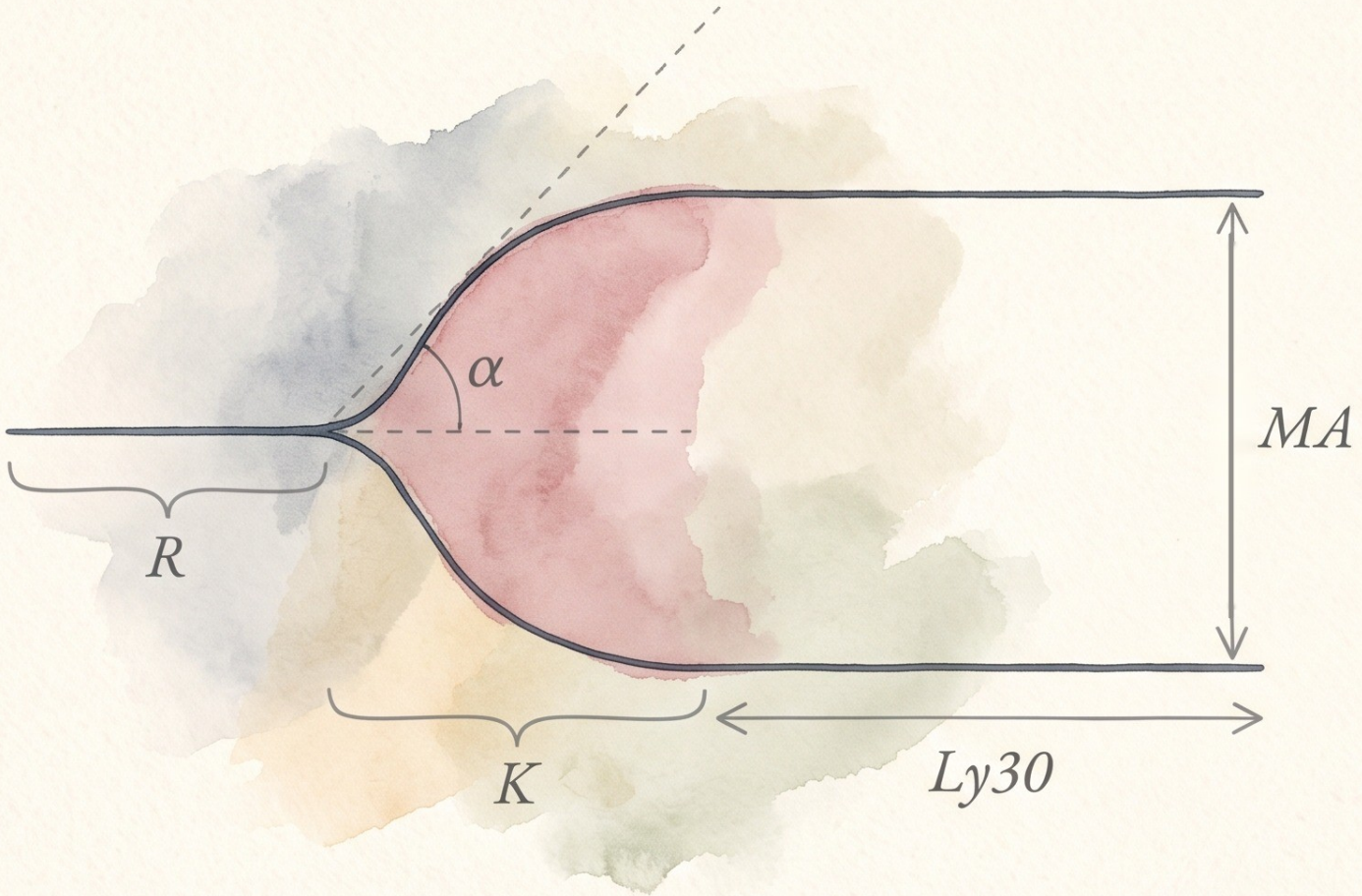
- **No plasma just to fix the number**
- The INR measures only part of clotting, and in ALF it doesn't predict who bleeds
- Plasma adds volume (lungs, brain) and erases the INR trend that prognosis and listing rely on

**What to do instead**

- Ultrasound-guided line, experienced operator, compressible site
- Vitamin K once; check fibrinogen and platelets; viscoelastic testing if available
- Treat bleeding. Plan high-risk procedures (like an ICP monitor) separately

*Update 2022 [PMID: 36001996]; SCCM 2020 [PMID: 32058387]*

# Is her blood thin?

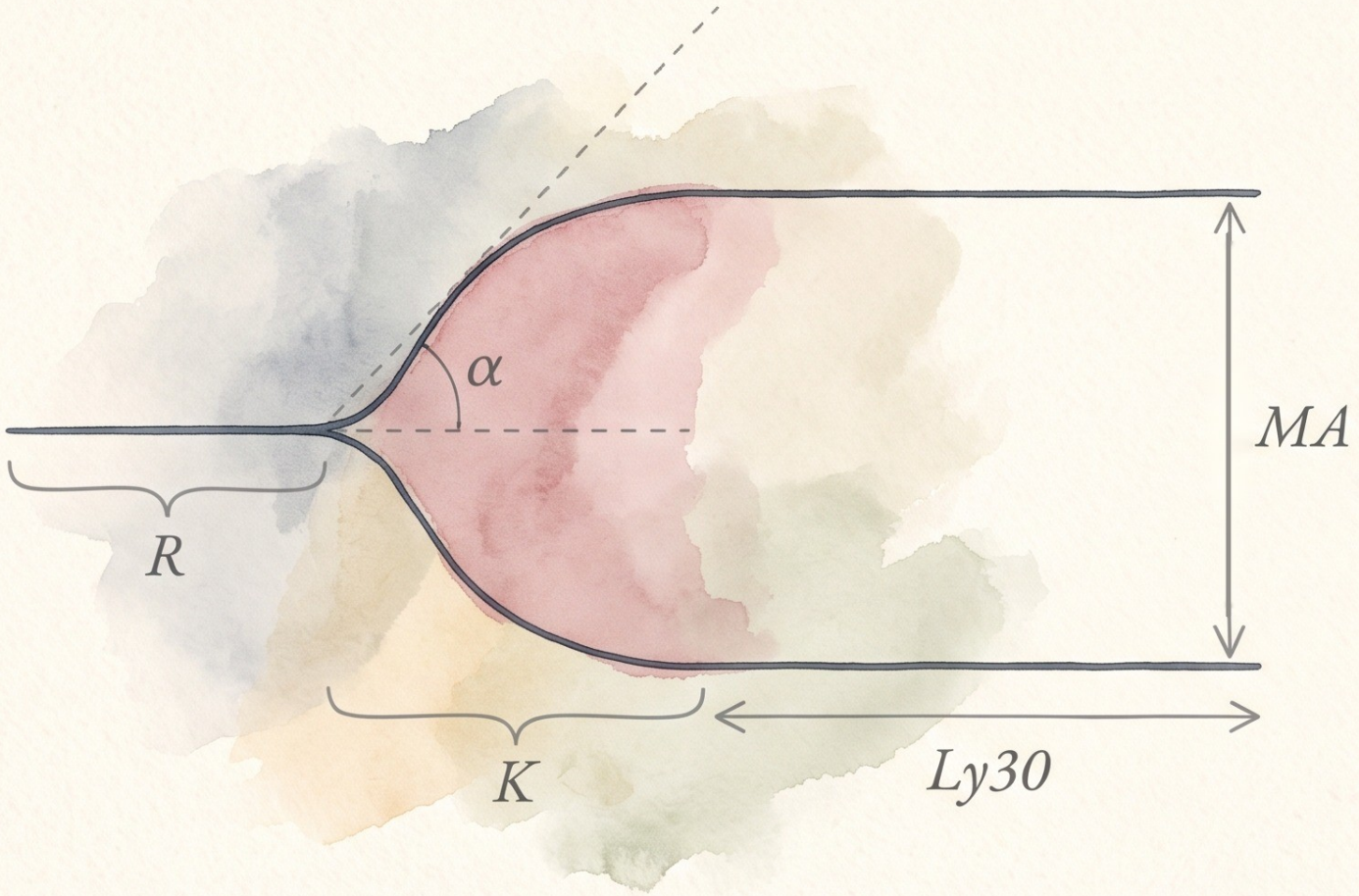



**What a TEG measures in whole blood**

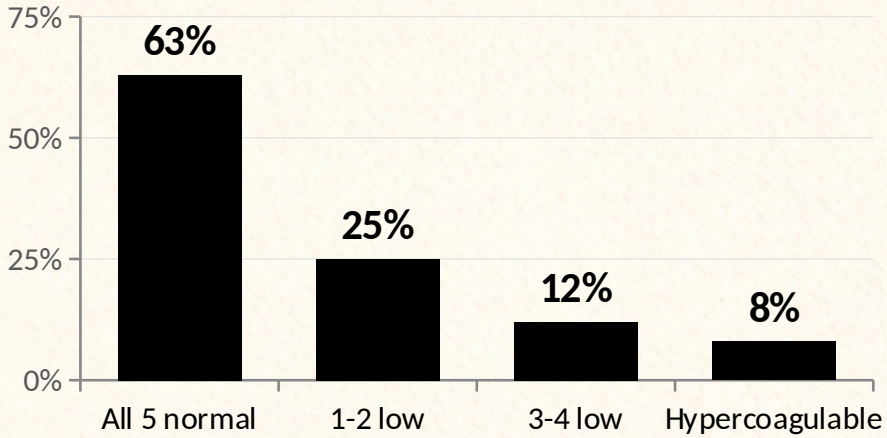
- **R:** time until the clot starts (clotting factors)
- **K and α-angle:** how fast the clot builds (fibrinogen)
- **MA:** how strong the clot gets (platelets, fibrinogen)
- **Ly30:** how much of the clot breaks down in 30 minutes

**Her INR is 4.1. What will her TEG show?**

# Is her blood thin?




51 patients, mean INR 3.4




- Most had a completely normal TEG, even with INRs up to 9.6
- A weak clot (low MA) only when platelets were below about 126,000
- Encephalopathy: stronger clots, not weaker

**A high INR doesn't mean she will bleed**

*Stravitz RT et al. J Hepatol 2012;56:129-36 [PMID: 21703173]*

# Rebalanced hemostasis

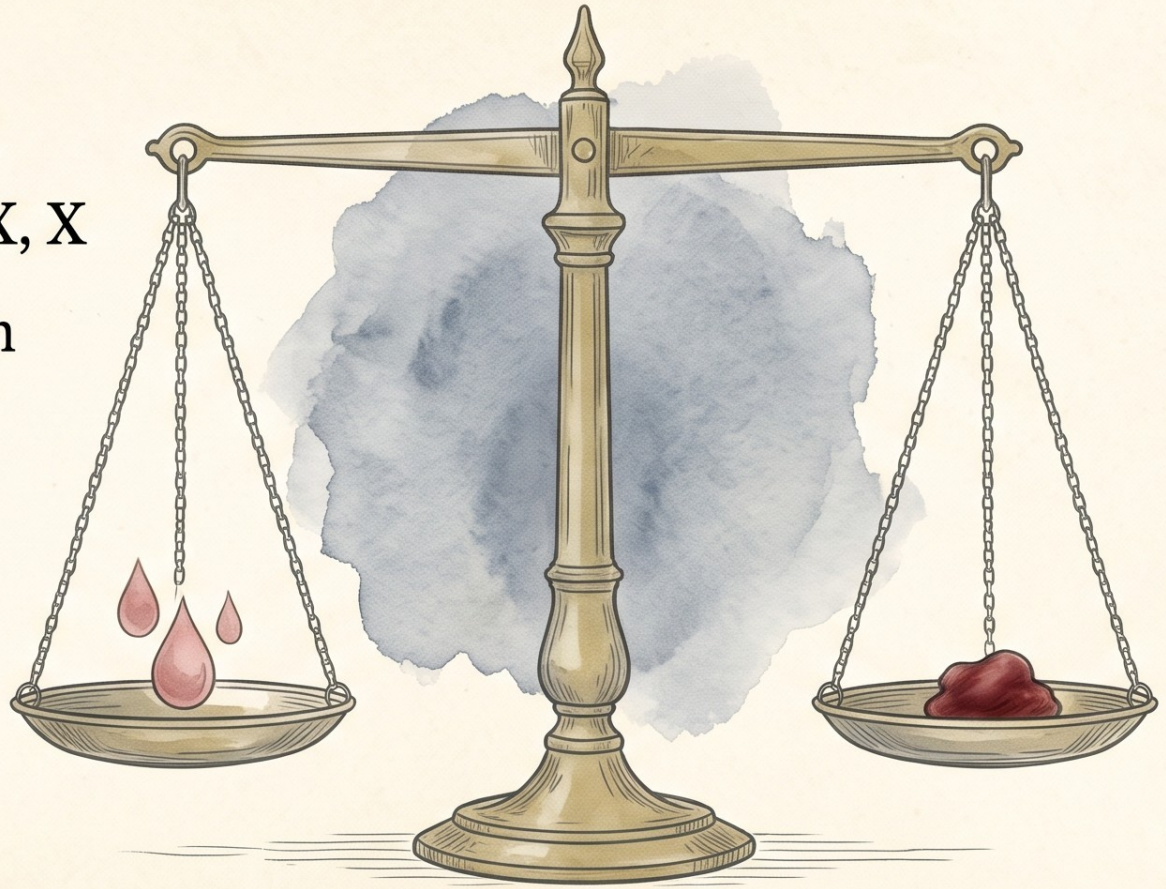
## Favors bleeding

Low factors II, V, VII, IX, X

Low fibrinogen when severe

Low platelets

The INR sees part of this side



## Favors clotting

Low protein C and protein S

Low antithrombin

High factor VIII and von Willebrand factor

The INR doesn't see this side

*Balanced in most patients, but easily tipped either way: both bleeding and thrombosis happen*

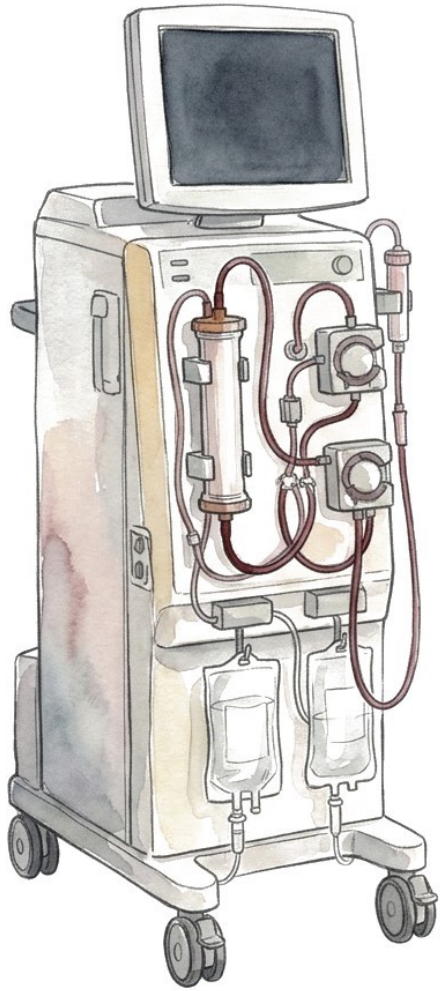
*Concept from Lisman T, Porte RJ. Blood 2010*

# Calling nephrology

**What do you ask them for?**

# Calling nephrology

- Indication: ammonia 214 and grade III encephalopathy. The creatinine doesn't matter here
- Start now and keep it running. A clotted filter means the ammonia climbs back
- One sodium plan: the CRRT fluid and the hypertonic saline together
- Watch phosphate and potassium, and calcium if they use citrate



*ACG ALF 2023 [PMID: 37377263]; update 2022 [PMID: 36001996]*

*I know what you're thinking...*

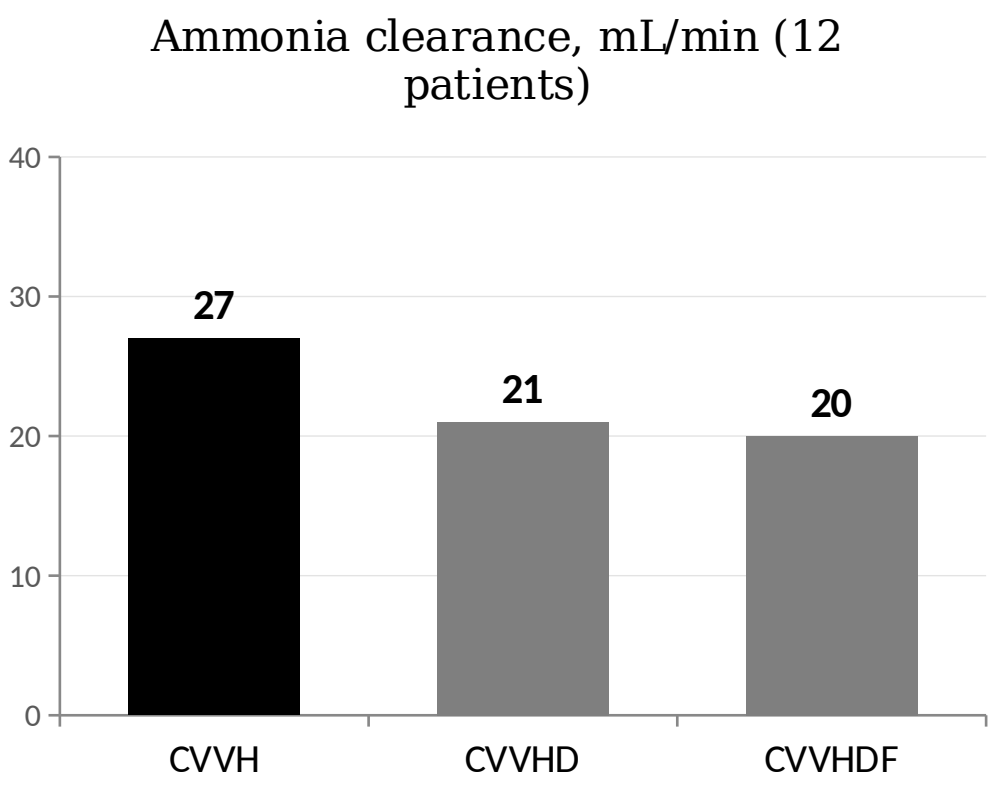
"CVVH doesn't clear ammonia"

**True or false?**


FALSE

# "CVVH doesn't clear ammonia"

- Ammonia is tiny (17 Da), water soluble and not protein bound
- It crosses the filter with the water (convection, CVVH) or down its gradient (diffusion, CVVHD)
- What matters is the effluent dose and hours on a working circuit, more than the mode
- More effluent, more clearance: **90 vs 35 mL/kg/h** (Slack 2014)
- 54 ALF patients on CRRT: median ammonia **151 → 52 µmol/L** by day 5 (Warrillow 2020)




**It does. About half the clearance of urea, so dose and filter life matter**

*Fisher C et al. Blood Purif 2022 [PMID: 35042216]; Warrillow SJ et al. Crit Care Med 2020 [PMID: 31939790]; Slack AJ et al. Liver Int 2014*

# Does early CRRT actually help?

| | US ALF Study Group (2018) | Pilot RCT (2026) |
|---|---|---|
| Design | Registry, observational | Open label, ALF with cerebral edema |
| Comparison | CRRT vs no CRRT | CRRT within 12 h vs delayed |
| Finding | | |
| Catch | | |

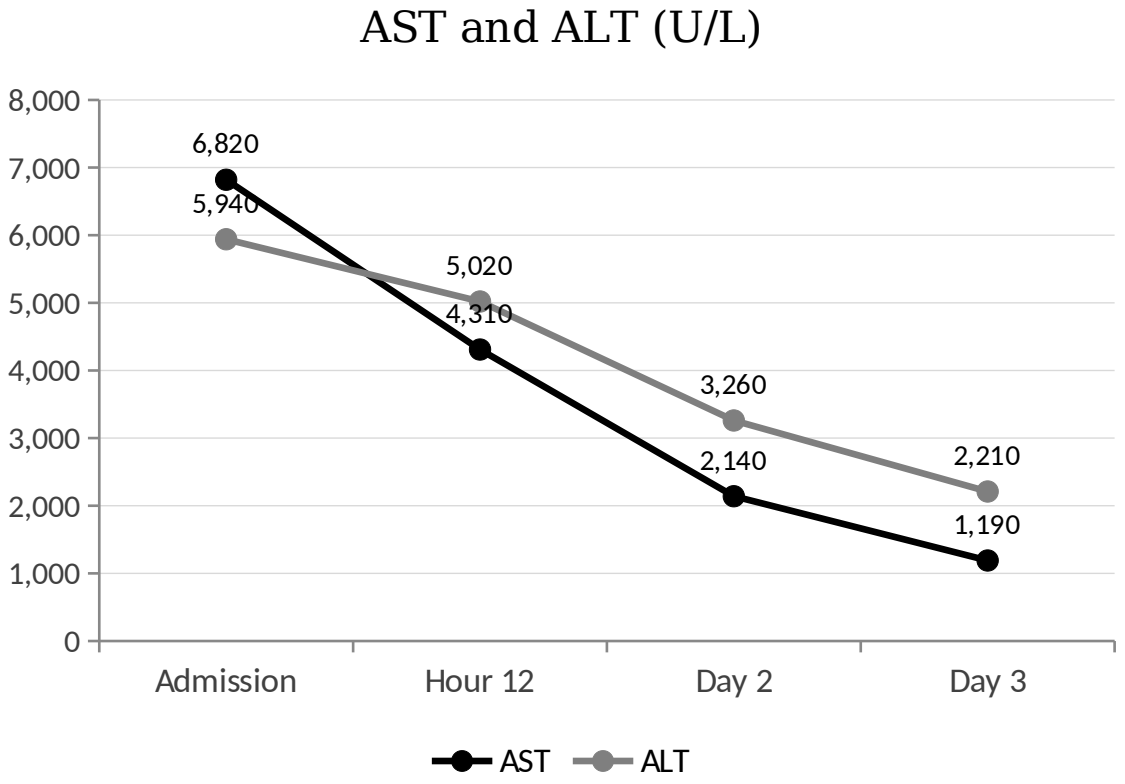
*Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; Aliment Pharmacol Ther 2026 [PMID: 42332991]*

# Does early CRRT actually help?

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|---|---|---|
| Design | Registry, observational | Open label, ALF with cerebral edema |
| Comparison | CRRT vs no CRRT | CRRT within 12 h vs delayed |
| Finding | Lower ammonia, better survival | Same 28-day survival; fewer deaths by day 7 |
| Catch | **Association; selection bias** | **Small, one center, protocol violations** |

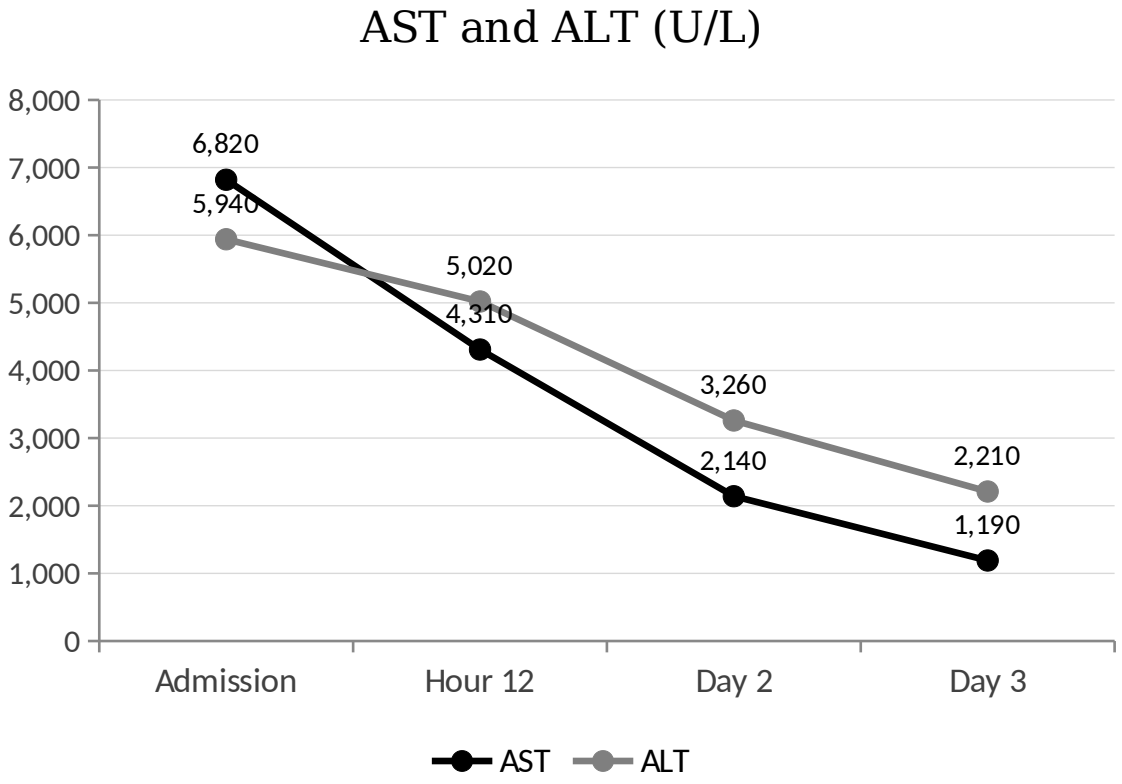
*Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; Aliment Pharmacol Ther 2026 [PMID: 42332991]*

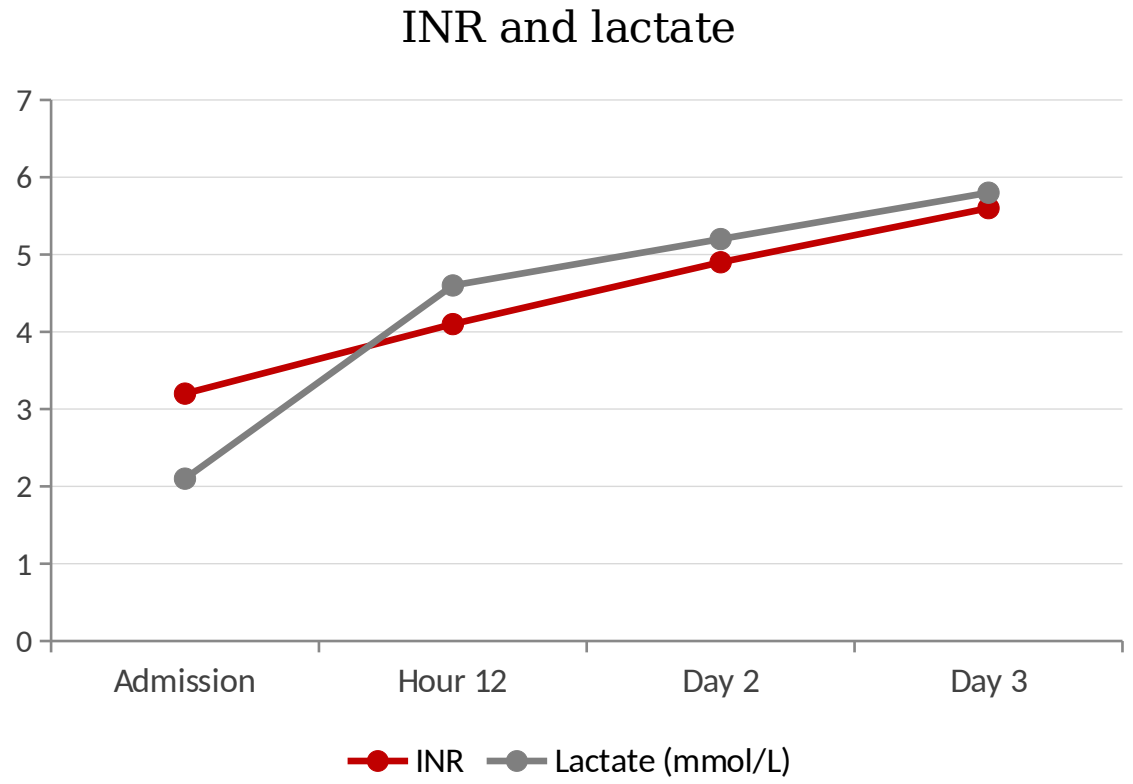
# Day 3: the enzymes are falling




**Is she getting better?**

# Day 3: the enzymes are falling




*Day 2 arterial pH 7.26 after fluids*

**No. The INR, the lactate and her brain are all worse**

*Made-up numbers for teaching*

# When do you call transplant hepatology?

**When she meets King's College criteria?**

**When MELD is high enough?**



# When do you call transplant hepatology?

**You should have called on admission, when you first suspected ALF**

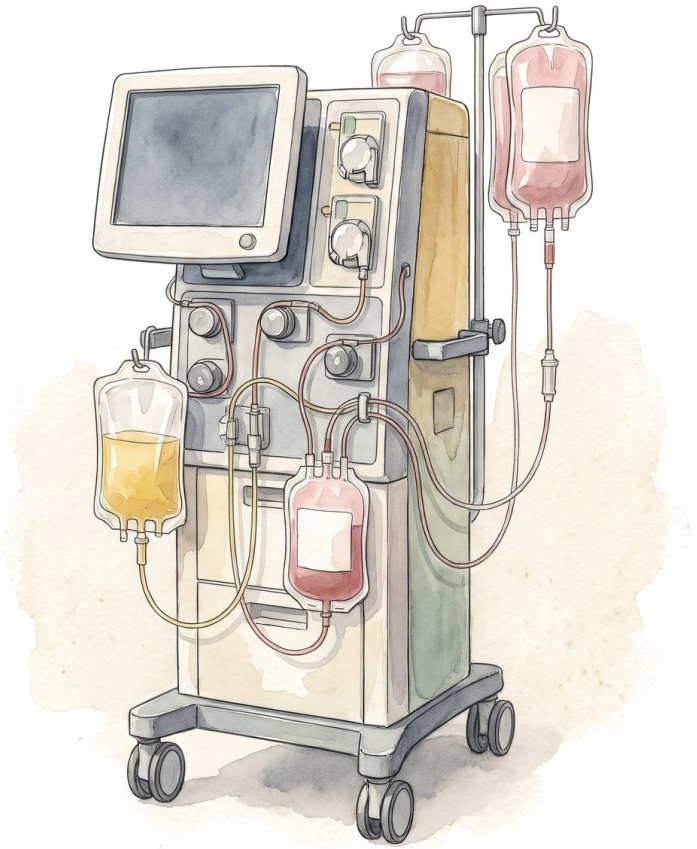
- King's College criteria predict death without transplant. They are specific, not sensitive
- They are not a referral threshold. If you wait for them, the evaluation starts too late
- Patient A: arterial pH **7.26** on day 2 after fluids. She now meets the acetaminophen criteria



*O'Grady 1989 criteria, as summarized in EASL 2017 [PMID: 28417882]; full criteria in the appendix*

# Does plasma exchange help in ALF?

| | Larsen 2016 | Burke 2025 |
|---|---|---|
| **Design** | RCT, 182 ALF | UK, 7 centers, 378 ALF |
| **Plasma exchange** | High volume × 3 days | 120 patients, matched |
| **Survival** | 58.7% vs 47.8% (HR 0.56) | 51.4% vs 62.6% (p = 0.12) |
| **Catch** | **Not blinded; no ACLF** | **Retrospective; sicker patients got it** |



**Better blood pressure, no proven survival benefit**

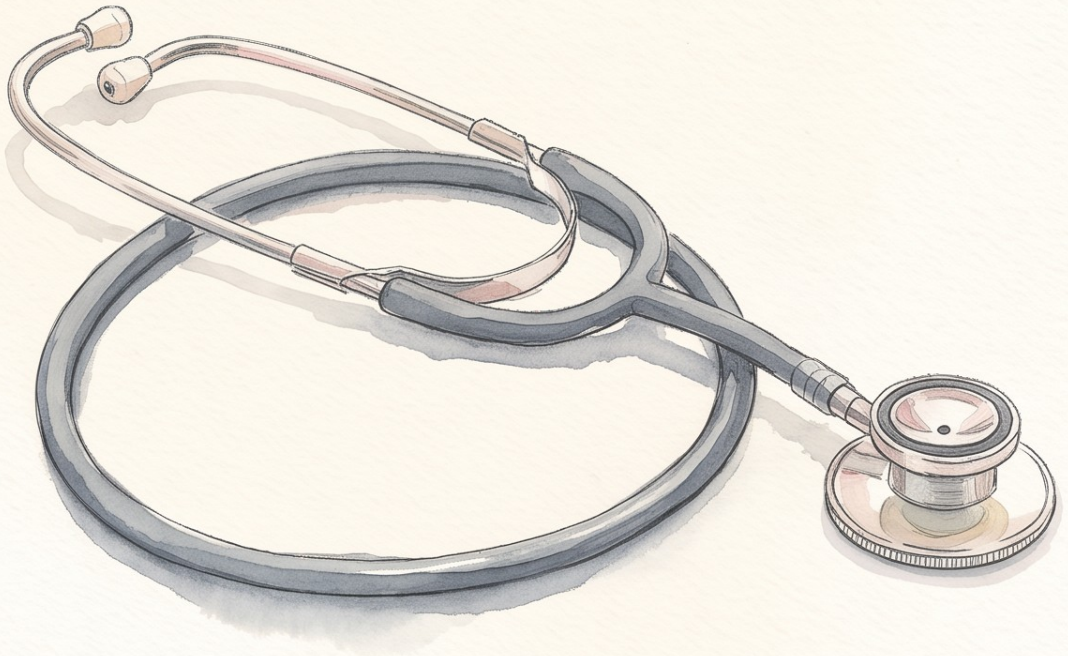
*Larsen FS et al. J Hepatol 2016 [PMID: 26325537]; Burke L et al. J Hepatol 2025;82:615-21 [PMID: 39362282]*

# Bed 4 on day 3

| Problem | What she's on | What we watch |
|---|---|---|
| Acetaminophen ALF, pH 7.26 | NAC; listed 1A on day 2 | Time to an offer |
| HE III-IV, ammonia 182 | Intubated, CO2 normal, Na 145-155, CRRT | Pupils, ammonia, Na |
| Lactate 5.8 and rising | Volume, MAP, look for infection | Lactate, MAP, pressors |
| INR 5.6, not bleeding | **No plasma** | Bleeding, INR trend |
| Infection risk | Surveillance cultures | Low bar for antimicrobials |

*Made-up numbers for teaching*

# Any questions so far?



# Bed 7

56M, cirrhosis and fever



# 56M, cirrhosis and fever

- Alcohol-related cirrhosis, ascites on diuretics; last drink 3 weeks ago
- Temp 38.4 °C, HR 112, BP 82/46
- RR 26, SpO2 92% on a 35% Venturi mask; PaO2 74
- Confused, asterixis (grade II)
- Urine output 15 mL/h

| | Now | Baseline |
|---|---|---|
| Bilirubin (mg/dL) | 5.8 | 2.1 |
| INR | 2.1 | 1.6 |
| Creatinine (mg/dL) | 2.6 | 0.9 |
| Sodium (mmol/L) | 127 | 133 |
| WBC (/µL) | 16,200 | |
| Platelets (/µL) | 62,000 | |
| Lactate (mmol/L) | 3.9 | |

**What's your next step?**

# 56M, cirrhosis and fever

- Alcohol-related cirrhosis, ascites on diuretics; last drink 3 weeks ago
- Temp 38.4 °C, HR 112, BP 82/46
- RR 26, SpO2 92% on a 35% Venturi mask; PaO2 74
- Confused, asterixis (grade II)
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| Sodium (mmol/L) | 127 | 133 |
| WBC (/µL) | 16,200 | |
| Platelets (/µL) | 62,000 | |
| Lactate (mmol/L) | 3.9 | |

**What's your next step?**

**Find the source: cultures, antibiotics**

What tipped him over?

**INR 2.1, platelets 62,000. Do you tap him?**



# What tipped him over?

- **Yes, now. No plasma, no platelets**
- Ascitic fluid: **PMN 1,240/mm³**. Spontaneous bacterial peritonitis

**Common triggers**

- Infection: SBP, pneumonia, urine, lines, C. difficile
- Alcohol-associated hepatitis
- GI bleeding
- Drugs: NSAIDs, sedatives, too much diuretic
- HBV flare, ischemia
- No trigger is found in some patients. Repeat the search if he gets worse



*EASL ACLF 2023 [PMID: 37364789]; SCCM 2023 [PMID: 37052436]*

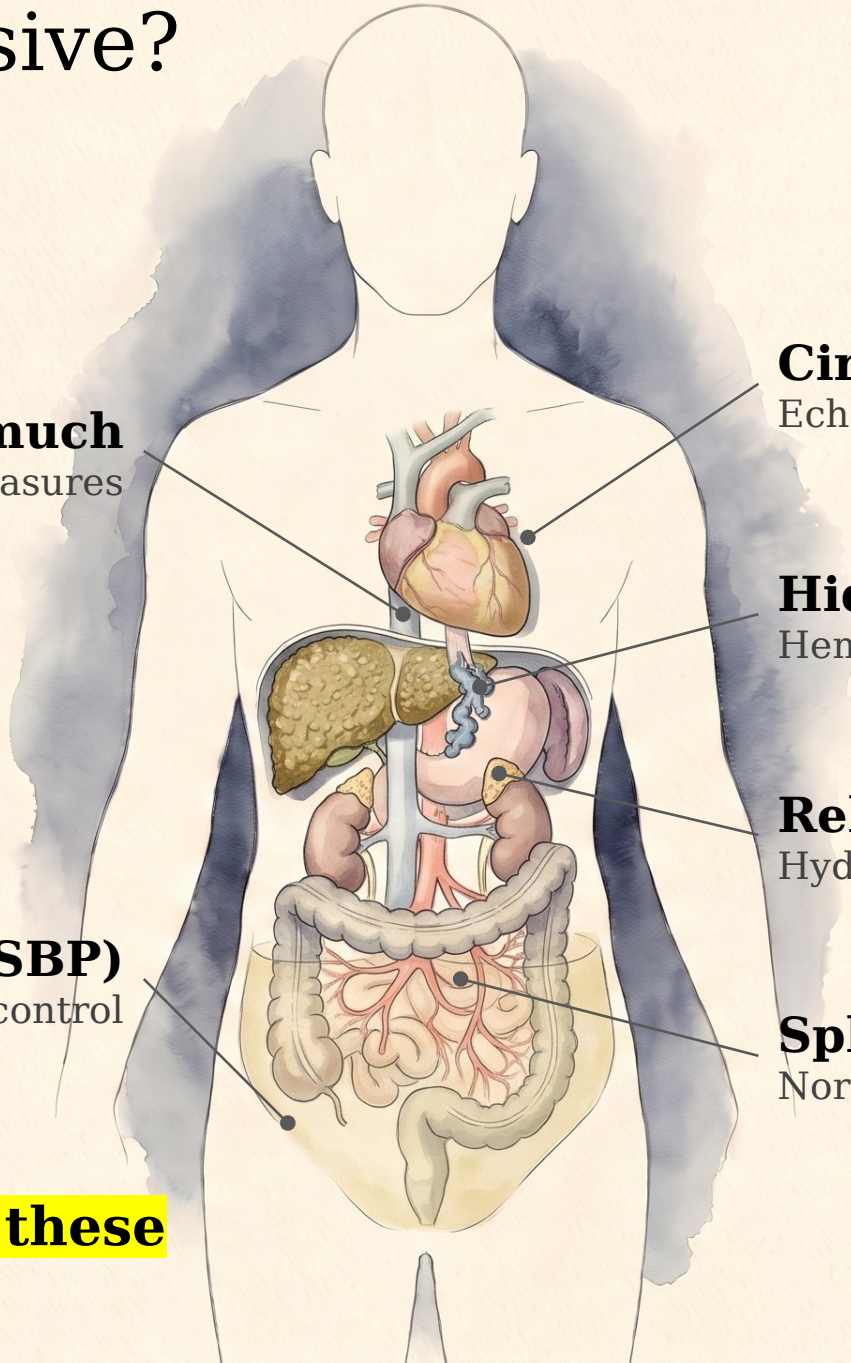
# Why is he hypotensive?



# Why is he hypotensive?

**Too little volume, or too much**
Ultrasound and dynamic measures

**Sepsis (SBP)**
Cultures, antibiotics within 1 h, source control



**Cirrhotic cardiomyopathy**
Echo

**Hidden bleeding**
Hemoglobin, rectal exam, think varices

**Relative adrenal insufficiency**
Hydrocortisone if shock is refractory

**Splanchnic vasodilation**
Norepinephrine, MAP 65, adjust to perfusion

**Usually more than one of these**

# ALF, decompensation or ACLF?

| | Acute liver failure | Acute decompensation | ACLF |
|---|---|---|---|
| **Liver before** | | | |
| **Defined by** | | | |
| **Usually kills** | | | |
| **ICU focus** | | | |
| **Transplant** | | | |

*EASL ALF 2017 [PMID: 28417882]; EASL ACLF 2023 [PMID: 37364789]*

# ALF, decompensation or ACLF?

| | Acute liver failure | Acute decompensation | ACLF |
|---|---|---|---|
| **Liver before** | No cirrhosis | Cirrhosis | Cirrhosis or chronic liver disease |
| **Defined by** | INR ≥ 1.5 + HE | Ascites, bleeding, HE, jaundice | Decompensation + organ failure |
| **Usually kills** | Brain, sepsis | Bleeding, infection | Infection, shock, organ failure |
| **ICU focus** | Brain, ammonia, early CRRT | Treat the trigger | Source, shock, kidney vs lung |
| **Transplant** | Urgent listing | Refer by MELD or complication | Narrow window; reassess daily |

*EASL ALF 2017 [PMID: 28417882]; EASL ACLF 2023 [PMID: 37364789]*

# Count the failing organs

| Organ | Failure (EASL-CLIF) | Patient B |
|---|---|---|
| Liver | Bilirubin ≥ 12 mg/dL | |
| Kidney | Creatinine ≥ 2.0 mg/dL or on RRT | |
| Brain | Encephalopathy grade III-IV | |
| Coagulation | INR ≥ 2.5 | |
| Circulation | Vasopressors (or terlipressin) | |
| Lungs | P/F ≤ 200, S/F ≤ 214, or ventilated (not for HE) | |

*CLIF-C organ failure score, EASL ACLF 2023 [PMID: 37364789]; CANONIC [PMID: 23474284]*

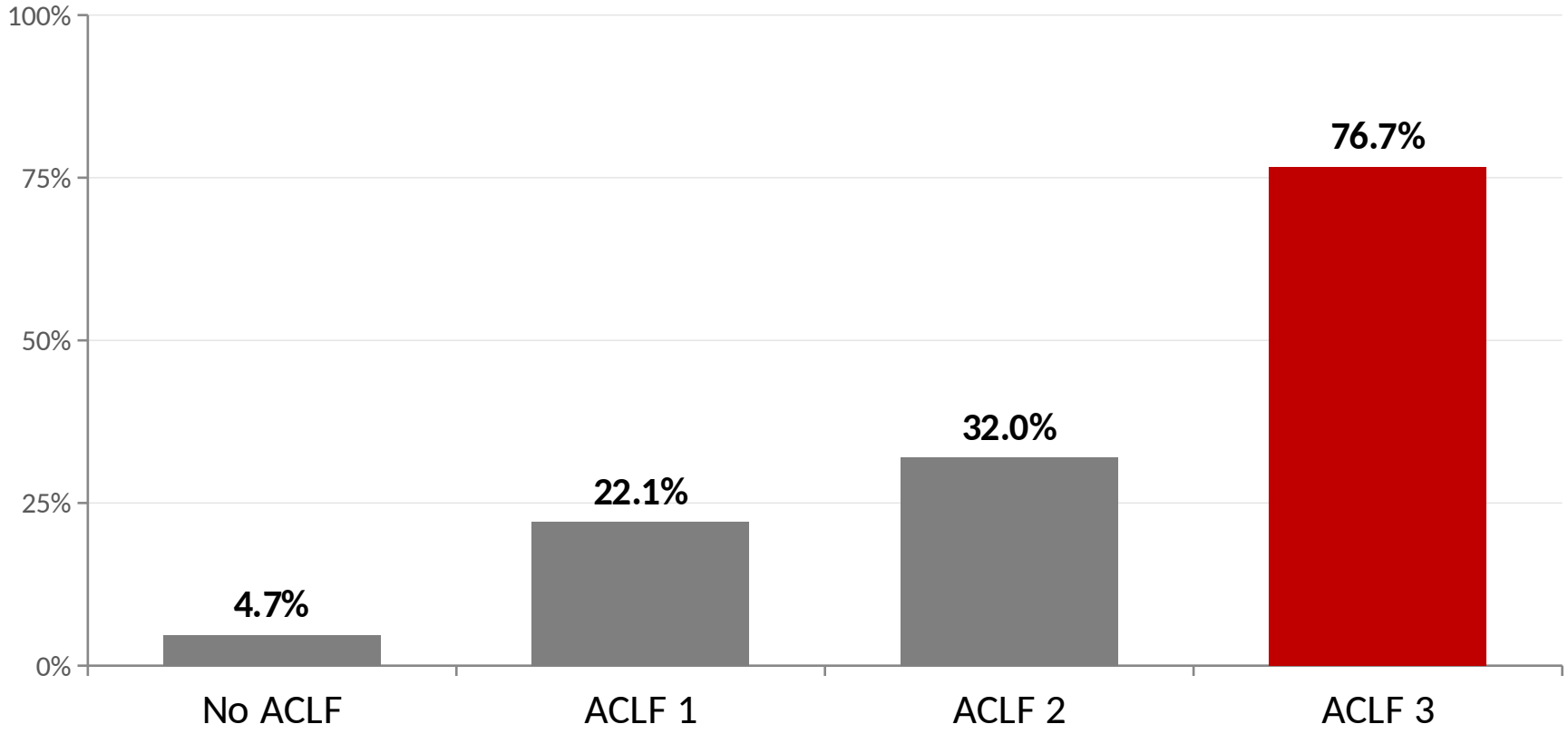
# Count the failing organs

| Organ | Failure (EASL-CLIF) | Patient B |
|---|---|---|
| Liver | Bilirubin ≥ 12 mg/dL | 5.8: no |
| Kidney | Creatinine ≥ 2.0 mg/dL or on RRT | **2.6: yes** |
| Brain | Encephalopathy grade III-IV | Grade II: no |
| Coagulation | INR ≥ 2.5 | 2.1: no |
| Circulation | Vasopressors (or terlipressin) | **Norepinephrine: yes** |
| Lungs | P/F ≤ 200, S/F ≤ 214, or ventilated (not for HE) | P/F 211: no |

**Two organ failures: ACLF grade 2**

*CLIF-C organ failure score, EASL ACLF 2023 [PMID: 37364789]; CANONIC [PMID: 23474284]*

# 28-day mortality by ACLF grade



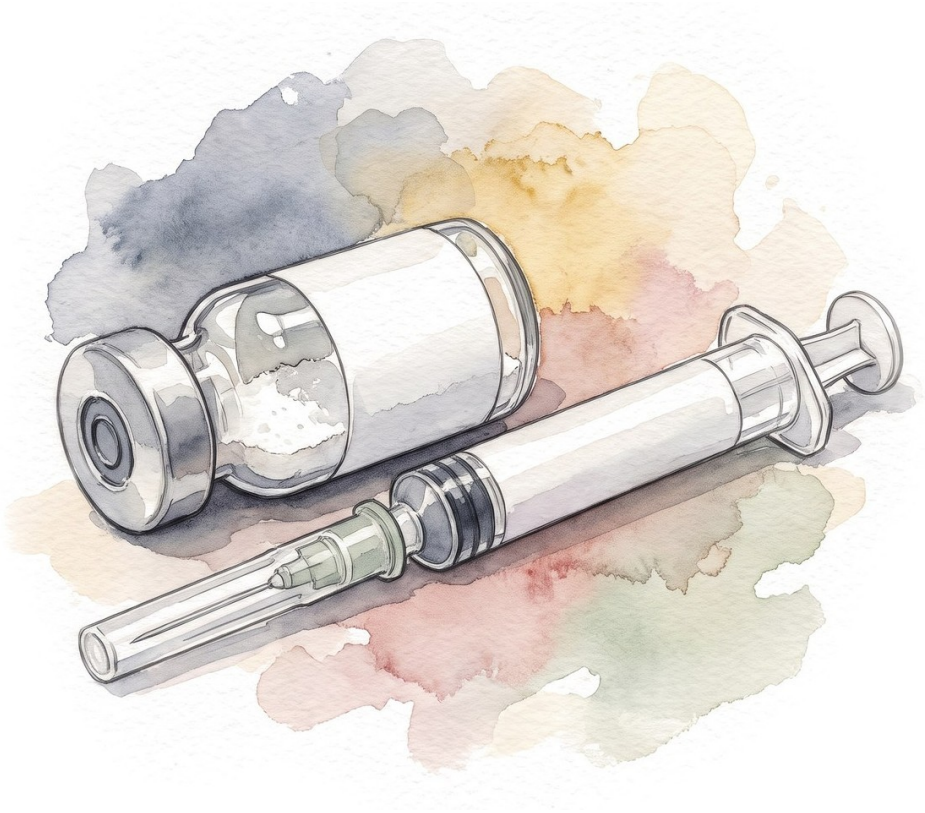

*CANONIC, 1,343 patients, 2013*

*Moreau R et al. Gastroenterology 2013 [PMID: 23474284]*

# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

**HRS? Start terlipressin?**



# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

| **HRS-AKI (ADQI-ICA 2024)** | **Bed 7, day 1** | |
|---|---|---|
| Cirrhosis with ascites | | |
| AKI by creatinine or urine output | | |
| No response to 24 h of volume, if needed | | |
| No other main cause (shock, toxins, structural) | | |

**Go down the list for him. Which boxes does he tick?**

*ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]*

# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

| **HRS-AKI (ADQI-ICA 2024)** | **Bed 7, day 1** | |
|---|---|---|
| Cirrhosis with ascites | Yes | ✅ |
| AKI by creatinine or urine output | Cr 0.9 → 3.0, urine 10 mL/h | ✅ |
| No response to 24 h of volume, if needed | 2 L on arrival; too early | ❌ |
| No other main cause (shock, toxins, structural) | **Septic shock on norepinephrine** | ❌ |

**Not HRS yet. Treat the shock first**

*ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]*

# Day 2: worse kidney, worse lungs

| | Admission | Day 2 |
|---|---|---|
| Albumin received | | 110 g (SBP dose, 1.5 g/kg at 73 kg) + 2 L crystalloid |
| Breathing | SpO2 92%, Venturi 35% | **SpO2 86% on Venturi 50%, RR 32, crackles** |
| Ultrasound | | Diffuse B-lines, IVC full |
| Creatinine (mg/dL) | 2.6 | 3.1 |
| Urine output | 15 mL/h | 10 mL/h |
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 |

**More albumin? Terlipressin? Something else?**

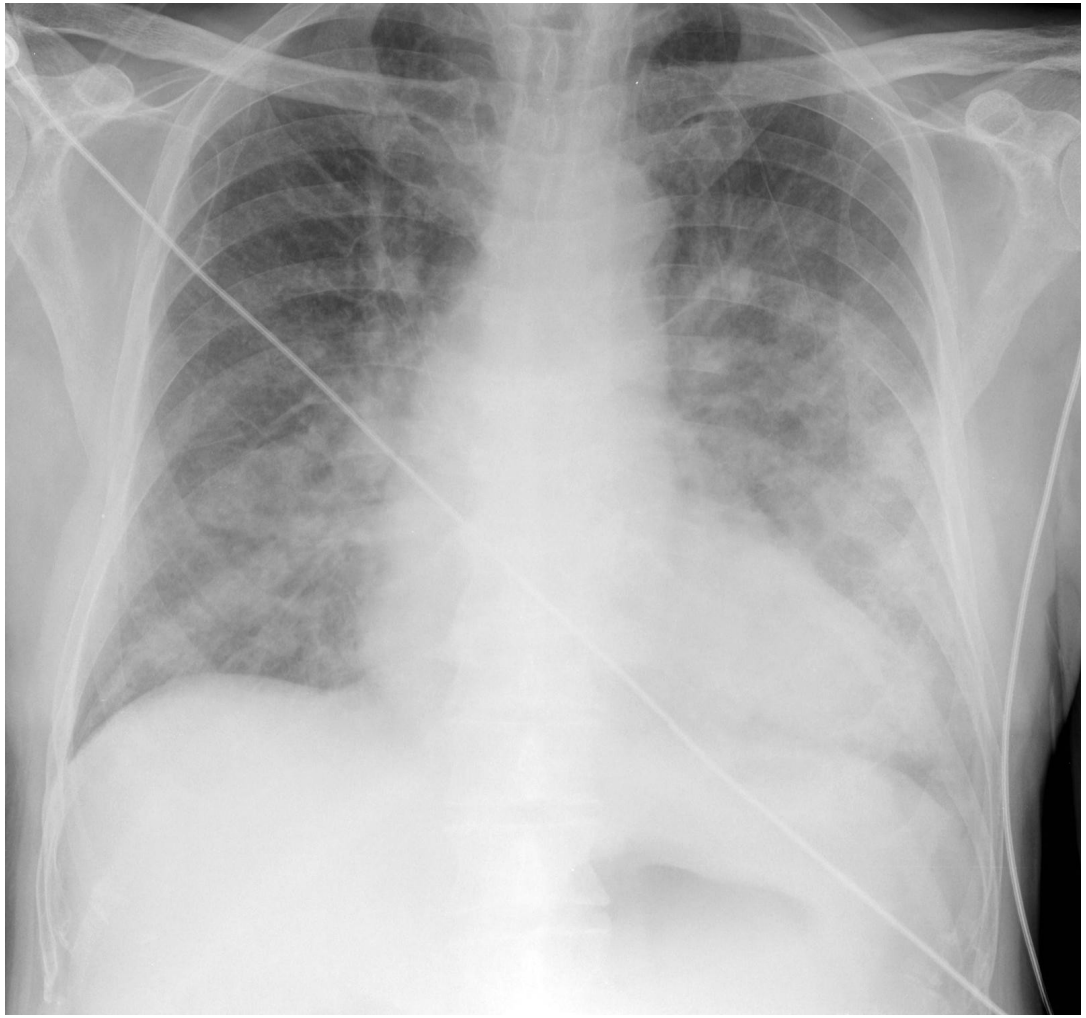
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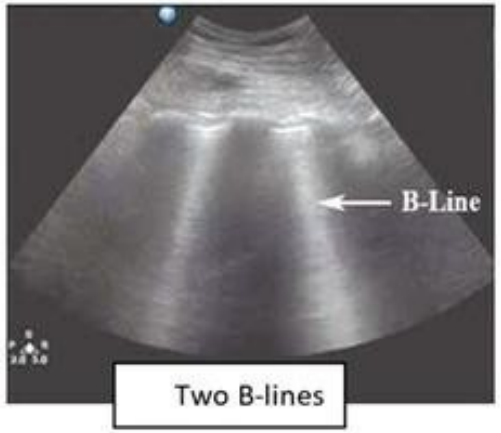
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| Urine output | 15 mL/h | 10 mL/h |
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 |

## First, look at his oxygen

**Now on high-flow, FiO2 0.6, SpO2 90%: SpO2/FiO2 150. That's a third organ failure**

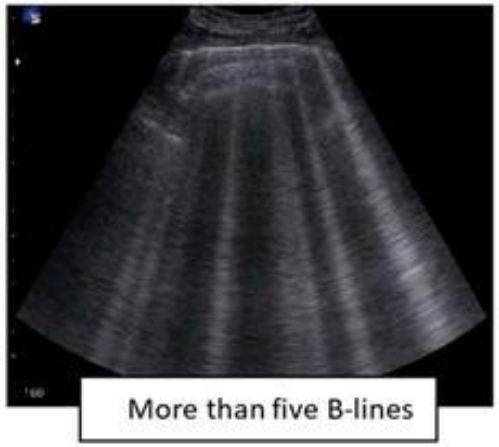
# Day 2: what do his lungs look like?

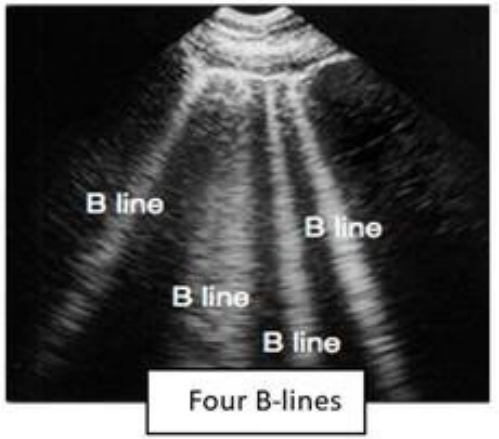








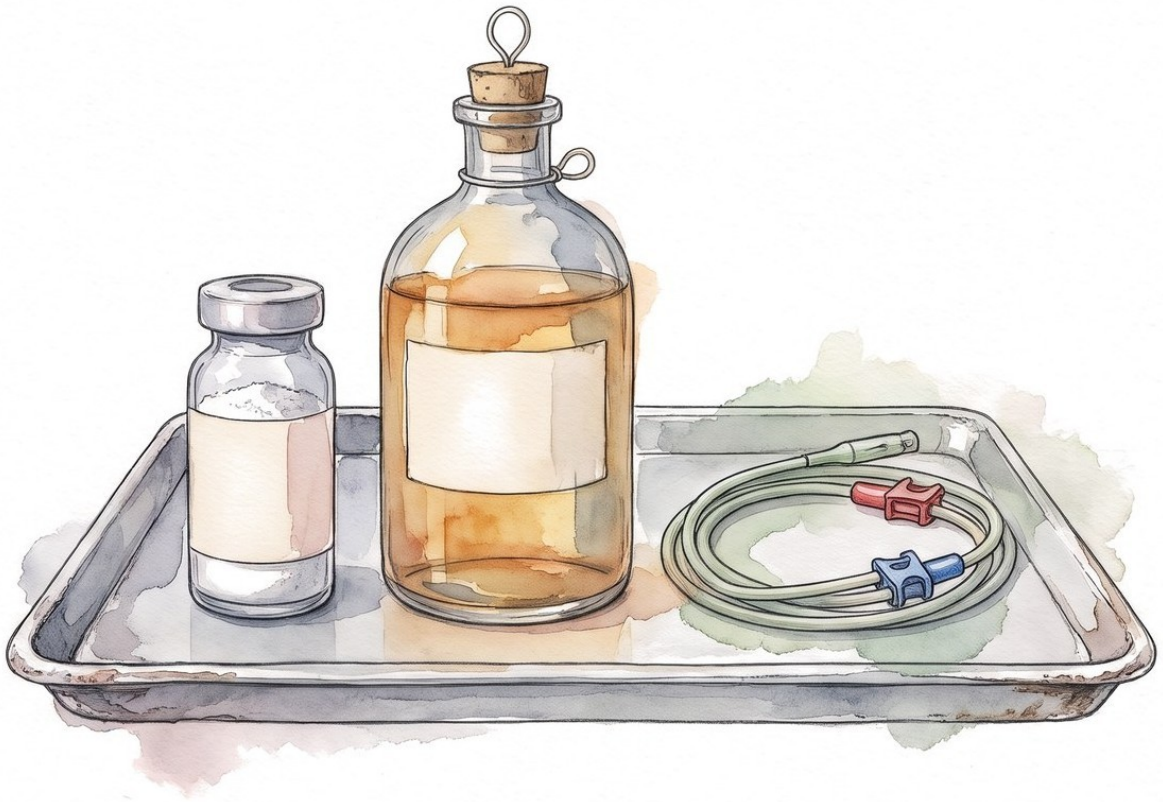



*Images for teaching, not this patient*

What would you do for his kidney today?

**Terlipressin, more albumin, or dialysis?**



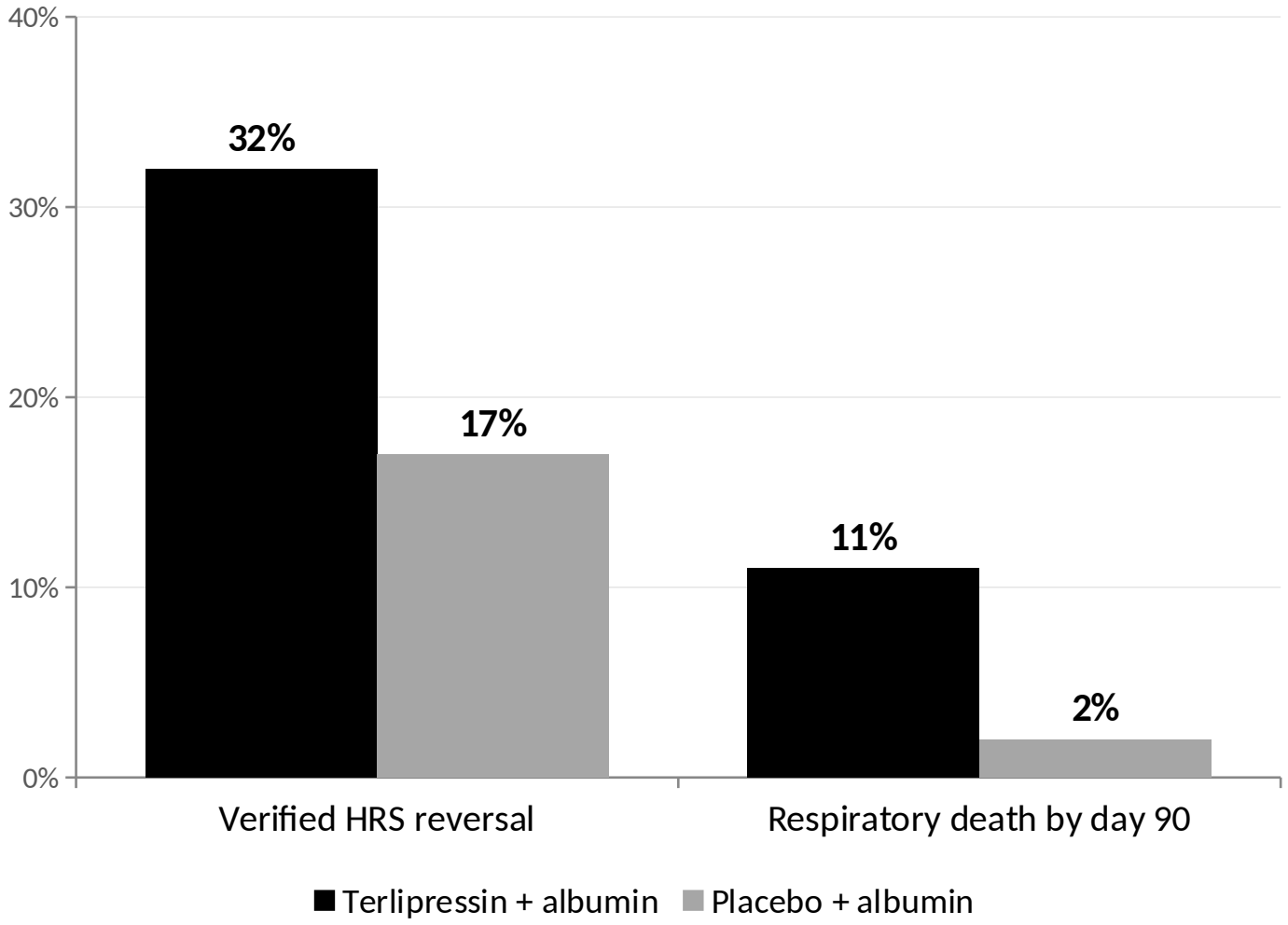
# What would you do for his kidney today?

- **Terlipressin: not for him today**
  - Label: don't start if SpO2 < 90% or breathing is getting worse
  - Label: avoid in ACLF grade 3 and in volume overload
- Stop the albumin while he is wet
- He is already on a vasoconstrictor. Norepinephrine, titrated to perfusion
- RRT if he stays overloaded. For HRS, RRT is a bridge to transplant: if he isn't a candidate, talk about goals before you start

**Check oxygen and volume before any kidney treatment**

*TERLIVAZ prescribing information, revised 6/2026; ADQI-ICA 2024 [PMID: 38527522]*

# Does HRS reversal mean survival?




**CONFIRM: type 1 HRS, 300 patients**

- More kidneys recovered
- **More patients died of respiratory failure**
- All-cause 90-day mortality was not different
- Respiratory failure as an adverse event: about 14% vs 5% (label)

*Wong F et al. NEJM 2021 [PMID: 33657294]; TERLIVAZ prescribing information, revised 6/2026*

# Should we treat the albumin number?

| | |
|---|---|
| **Study** | 777 inpatients: albumin to ≥ 30 g/L vs standard care |
| **Finding** | Infection, kidney dysfunction or death: 29.7% vs 30.2% |
| **The catch** | **More serious adverse events, including pulmonary edema** |

**When albumin is indicated**

- SBP: 1.5 g/kg on day 1, 1 g/kg on day 3
- Paracentesis over 5 L: 6 to 8 g per liter removed
- HRS-AKI: with a vasoconstrictor, dosed to volume status

*China L et al. NEJM 2021 [PMID: 33657293]; ADQI-ICA 2024 [PMID: 38527522]*

# What changed since yesterday?

| | Day 1 | Day 2 | Day 3 | Day 5 |
|---|---|---|---|---|
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 | 0.12 | Off |
| Oxygen, SpO2 | Venturi 0.35, 92% | HFNC 0.6, 90% | HFNC 0.4, 95% | 2 L, 97% |
| Lung failure? | No (P/F 211) | Yes (S/F 150) | No (S/F 238) | No |
| Kidney | Cr 2.6 | Cr 3.1 | CRRT for fluid | On CRRT |
| Encephalopathy | Grade II | Grade II | Grade I | Grade I |
| Bilirubin / INR | 5.8 / 2.1 | 6.4 / 2.3 | 6.1 / 2.2 | 5.2 / 2.0 |
| ACLF grade | 2 | **3** | 2 | ? |

**Better or worse? What do you tell his family?**

*Made-up numbers for teaching. CLIF-C ACLF score [PMID: 24950482]; EASL ACLF 2023 [PMID: 37364789]*

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| Encephalopathy | Grade II | Grade II | Grade I | Grade I |
| Bilirubin / INR | 5.8 / 2.1 | 6.4 / 2.3 | 6.1 / 2.2 | 5.2 / 2.0 |
| ACLF grade | 2 | **3** | 2 | **1** |

- The grade at days 3 to 7 predicts outcome better than the grade at admission
- Persistent ACLF 3 at days 3 to 7, especially with four or more organ failures, has a very high mortality without transplant
- If he is not a transplant candidate, discuss goals of care with the family and involve palliative care

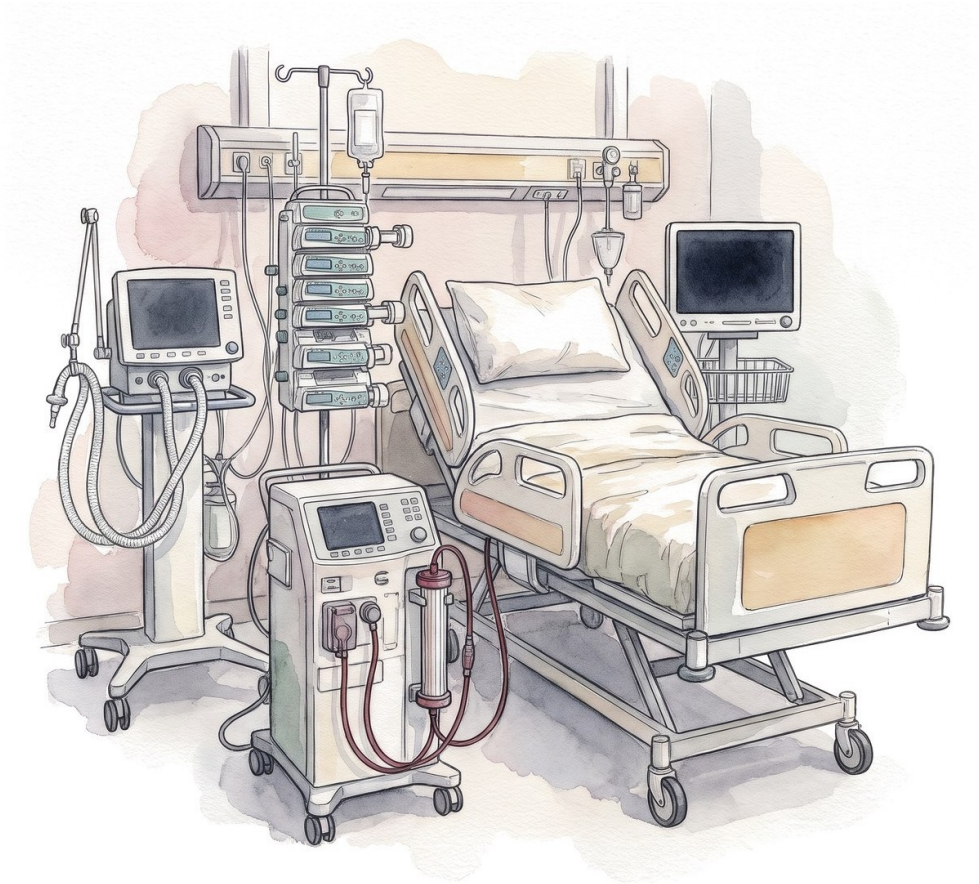
*Made-up numbers for teaching. CLIF-C ACLF score [PMID: 24950482]; EASL ACLF 2023 [PMID: 37364789]*

# Transplant in ACLF



Is he too sick for a transplant?

**ACLF 3 on a ventilator. Would you call?**



# Is he too sick for a transplant?

- A ventilator, pressors or dialysis do not by themselves exclude transplant
- Selected patients with ACLF grade 3 do well after transplant, if it happens early

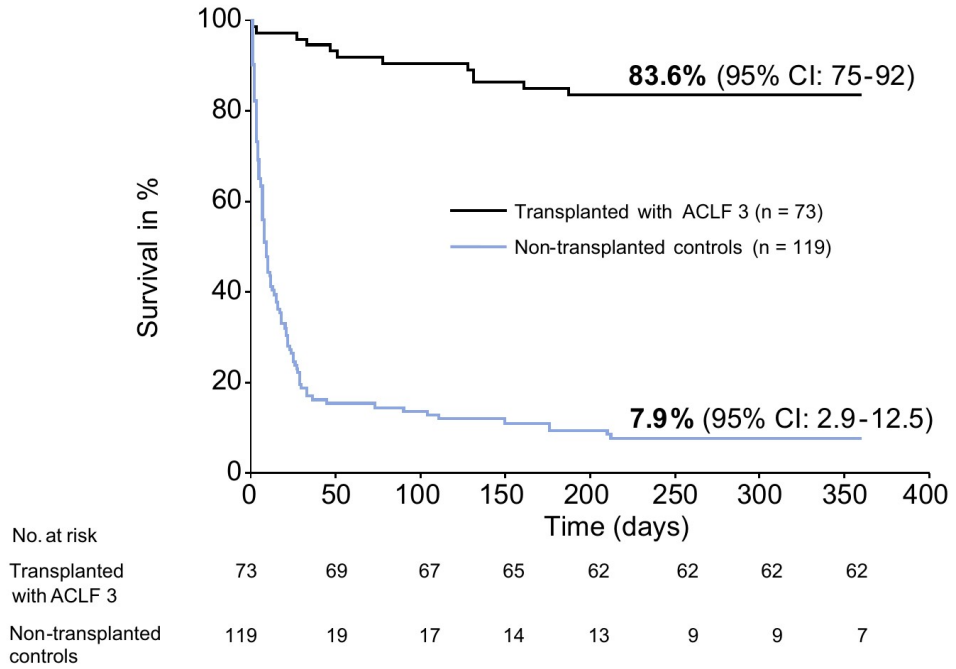
**What makes transplant unlikely**

- Uncontrolled infection
- Rising vasopressor needs, severe respiratory failure
- Irreversible brain injury
- Frailty and comorbidity
- Alcohol: centers now evaluate individually rather than by a fixed abstinence rule

Research Article

EASL JOURNAL OF HEPATOLOGY

**Liver transplantation in the most severely ill cirrhotic patients: A multicenter study in acute-on-chronic liver failure grade 3**




*Selected patients, not randomized*

*Artru F et al. J Hepatol 2017 [PMID: 28645736]; Lancet Gastroenterol Hepatol 2024 [PMID: 38309288]; AASLD/AST 2026 [PMID: 41405234]*

# What happened to them

## Bed 4



**Day 0**
Admitted, NAC

**Day 1**
Intubated, CRRT

**Day 2**
Listed 1A

**Day 4**
Transplanted

**Day 19**
Home

## Bed 7



**Day 1**
SBP, ACLF 2

**Day 2**
ACLF 3

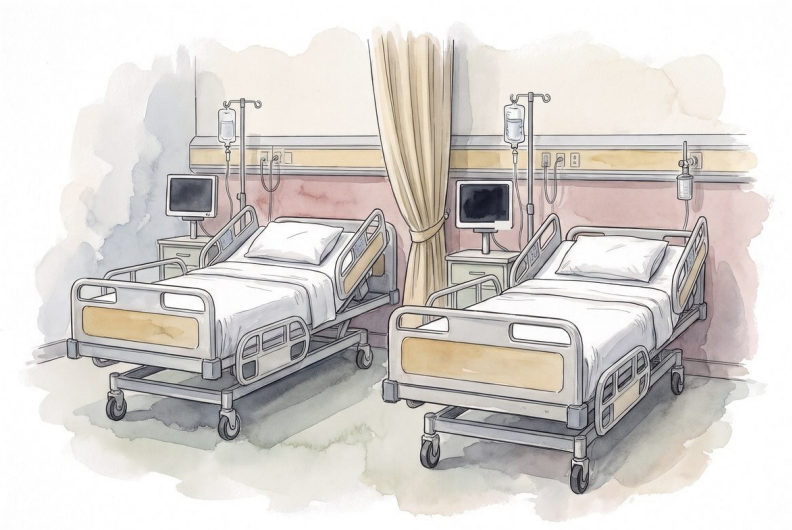
**Day 5**
Off pressors, ACLF 1

**Day 12**
Off CRRT

**Day 21**
Home, transplant clinic

*Made-up outcomes for teaching*

# Back to our patients

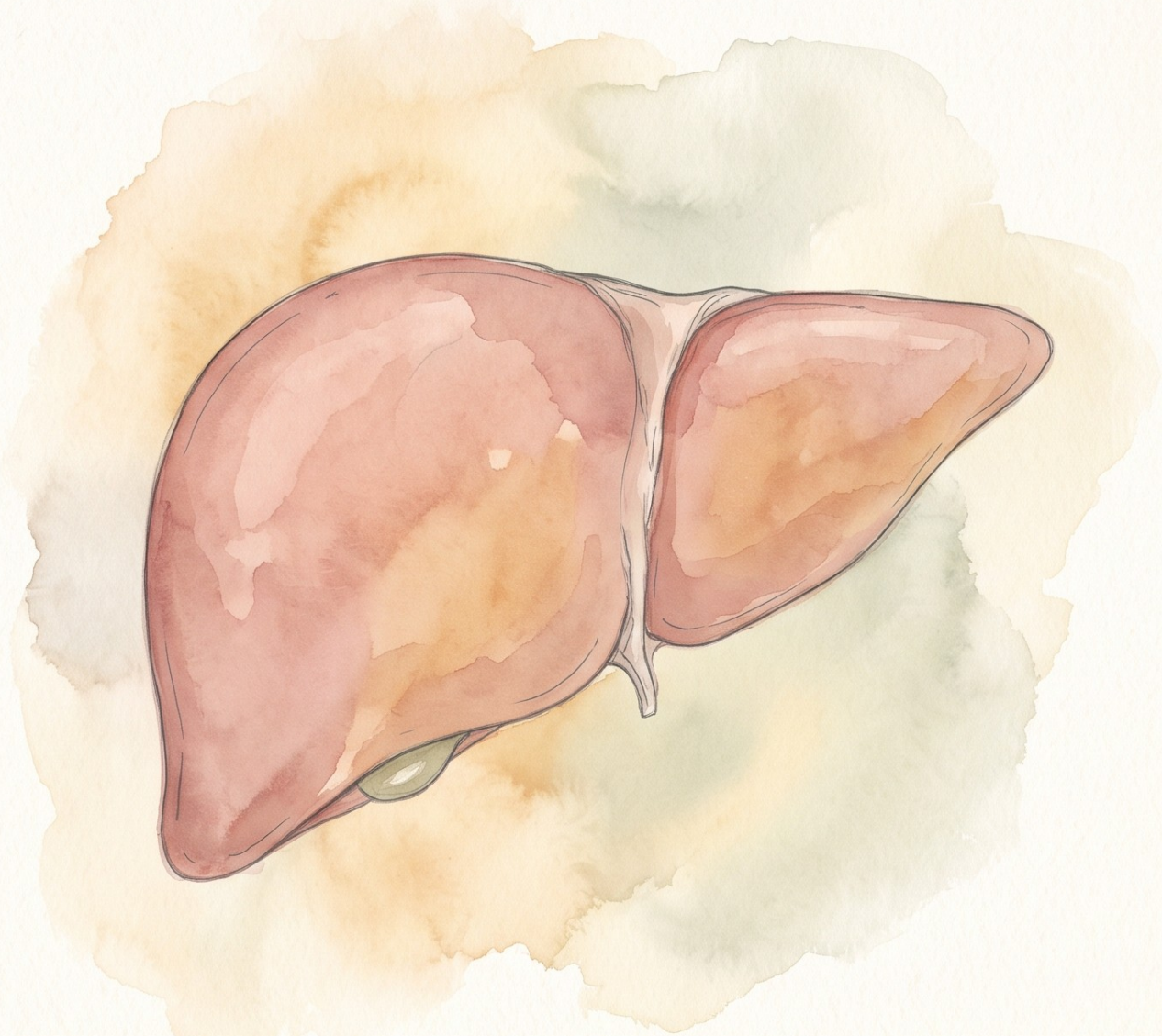


- Who needed transplant hepatology before the labs?
- Why did a falling ALT worry us more?
- When is CRRT right before the creatinine rises?
- Why doesn't cirrhosis lactulose data apply to ALF?
- Why not albumin and terlipressin for bed 7?
- What would make you change yesterday's plan?

# Summary

- High AST and ALT alone are not liver failure. Get an INR and examine the patient
- Stabilize, treat the cause and call transplant hepatology in the same hour
- ALF: **protect the brain**. Intubate at grade III, keep the CO2 normal, follow your center's sodium plan, and start CRRT for rising ammonia
- ACLF: count the organs, find the trigger
- ACLF: check oxygenation and volume status before albumin or terlipressin
- Reassess organ failures daily. The trend matters more than the admission score





# Questions?

Slides (PDF)

# Appendix: West Haven grades

| Grade | What you see |
|---|---|
| I | Short attention, mild confusion, reversed sleep, subtle personality change |
| II | Lethargy, disorientation to time, obvious personality change, asterixis |
| III | Somnolent but rousable, grossly disoriented, bizarre behavior |
| IV | Coma |

# Appendix: King's College criteria

| Acetaminophen | Not acetaminophen |
|---|---|
| **Arterial pH < 7.30 after fluids**<br>*or all three within 24 hours:*<br>INR > 6.5<br>Creatinine > 3.4 mg/dL<br>Grade III-IV encephalopathy<br>*(Modified: lactate > 3.5 at 4 h or > 3.0 at 12 h after fluids)* | **INR > 6.5**<br>*or any three of:*<br>Age < 10 or > 40<br>Drug reaction or unknown cause<br>Jaundice to encephalopathy > 7 days<br>INR > 3.5<br>Bilirubin > 17.5 mg/dL |

*Original O'Grady criteria; the lactate line is the later modification*

*O'Grady 1989 criteria, as summarized in EASL 2017 [PMID: 28417882] and update 2022 [PMID: 36001996]*

# Appendix: arterial ammonia studies in ALF

| Study (arterial, µmol/L) | Patients | What they found | The catch |
|---|---|---|---|
| Clemmesen 1999 | 44 ALF | **Herniated: mean 230 vs 118; all above 146** | Small, one center |
| Bhatia 2006 | 80 ALF | ≥ 124 on admission predicted death | Endpoint was death, not ICH |
| Bernal 2007 | 165, HE III-IV | > 100: severe HE. **> 200: ICH in 55%** | 200 missed most ICH |
| Kumar 2012 | 295 ALF | **≥ 122 for 3 days: survival 23% vs 72%** | Trend beat a single value |

**Above 150 to 200 and not falling: worry about the brain**

*ICH can occur below 200. Interpret the level with the HE grade and the trend*

*Clemmesen 1999 [PMID: 10051463]; Bhatia 2006 [PMID: 16024550]; Bernal 2007 [PMID: 17685471]; Kumar 2012 [PMID: 22521861]*

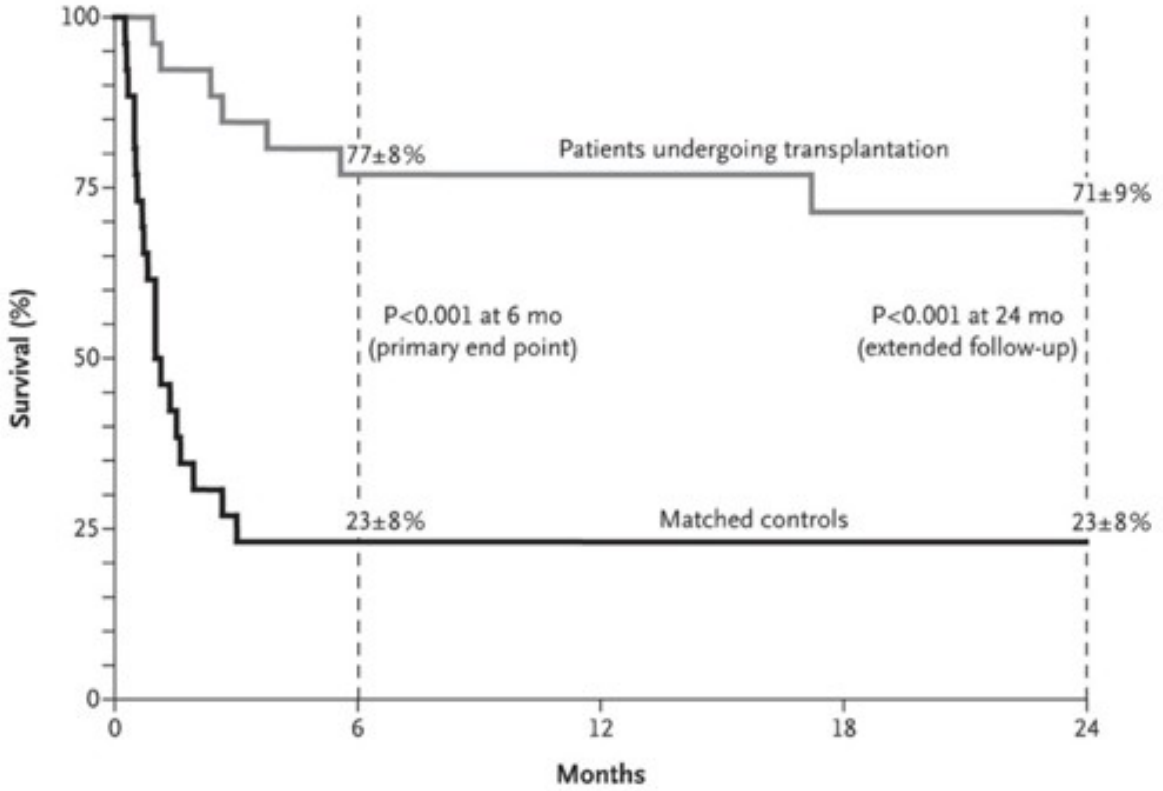
# Appendix: which ACLF?

| | EASL-CLIF | NACSELD | APASL |
|---|---|---|---|
| **Who** | Decompensated cirrhosis | Cirrhosis, inpatient | CLD ± cirrhosis |
| **Defined by** | Organ failures (CLIF-C) | ≥ 2 extrahepatic failures | Liver insult: jaundice + INR ≥ 1.5 |
| **Picks up** | Broader, earlier | Sicker patients | A primary liver insult |

**They find different patients. Say which one you're using**

*Kyoto consensus 2025 [PMID: 39961976]; ACG ACLF 2022 [PMID: 35006099]; AASLD 2024 [PMID: 37939273]*

# Appendix: early transplant for alcohol-associated hepatitis




| No. at Risk | | | | | |
|---|---|---|---|---|---|
| Patients undergoing transplantation | 26 | 20 | 15 | 14 | 13 |
| Matched controls | 26 | 6 | 6 | 5 | 4 |

- 26 patients with a first episode of severe alcohol-associated hepatitis, not responding to steroids
- Highly selected: first liver-related event, close family support, no severe other illness, agreed to abstinence
- 6-month survival **77% vs 23%**
- Recent drinking alone is not a reason to skip the call

*Mathurin P et al. N Engl J Med 2011 [PMID: 22070476]*

# Appendix: calling transplant hepatology

**Call first. Gather this while you wait for the call back**

| Patient A (ALF) | Patient B (ACLF) |
|---|---|
| Cause, days from jaundice to HE | Cause of cirrhosis, recent alcohol use |
| Encephalopathy grade, pupils, sedation | Precipitant and source control |
| Airway and ventilator settings | Organ failures today and yesterday |
| Ammonia, sodium, CRRT | Pressor dose, oxygen, RRT |
| INR, lactate, pH, glucose trends | MELD, CLIF-C scores |
| Pressors, infection | Frailty, comorbidities |
| Social support, intent of ingestion | Prior transplant evaluation, support |

# Appendix: sources (1)


- SCCM ALF/ACLF ICU guidelines, part 1. Crit Care Med 2020 [PMID: 32058387]
- SCCM ALF/ACLF ICU guidelines, part 2. Crit Care Med 2023 [PMID: 37052436]
- EASL acute liver failure guideline. J Hepatol 2017 [PMID: 28417882]
- ACG acute liver failure guideline. Am J Gastroenterol 2023 [PMID: 37377263]
- Management of acute liver failure: update 2022. Semin Liver Dis [PMID: 36001996]
- Acute liver failure: a practical update. JHEP Rep 2024 [PMID: 39170946]
- EASL ACLF guideline. J Hepatol 2023 [PMID: 37364789]
- AASLD ACLF and critically ill cirrhosis guidance. Hepatology 2024 [PMID: 37939273], errata [PMID: 38551831], [PMID: 42467893]
- ACG ACLF guideline. Am J Gastroenterol 2022 [PMID: 35006099]
- ADQI-ICA consensus on AKI in cirrhosis. J Hepatol 2024 [PMID: 38527522]
- Surviving Sepsis Campaign 2026. Crit Care Med [PMID: 41869847]
- EASL autoimmune hepatitis guideline. J Hepatol 2025 [PMID: 40348684]
- EASL hepatic encephalopathy guideline. J Hepatol 2022 [PMID: 35724930]
- US/Canada acetaminophen poisoning consensus. JAMA Netw Open 2023 [PMID: 37552484]

# Appendix: sources (2)


- Moreau R et al. CANONIC. Gastroenterology 2013 [PMID: 23474284]
- Jalan R et al. CLIF-C ACLF score. J Hepatol 2014 [PMID: 24950482]
- Lee WM et al. NAC in non-acetaminophen ALF. Gastroenterology 2009 [PMID: 19524577]
- Clemmesen JO et al. Cerebral herniation and arterial ammonia. Hepatology 1999 [PMID: 10051463]
- Bernal W et al. Arterial ammonia, HE and intracranial hypertension. Hepatology 2007 [PMID: 17685471]
- Bhatia V et al. Arterial ammonia in ALF. Gut 2006 [PMID: 16024550]
- Kumar R et al. Persistent hyperammonemia in ALF. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]
- Cardoso FS et al. CRRT and ammonia in ALF. Hepatology 2018 [PMID: 28859230]
- Pre-emptive CRRT in ALF with cerebral edema. Aliment Pharmacol Ther 2026 [PMID: 42332991]
- Larsen FS et al. High-volume plasma exchange in ALF. J Hepatol 2016 [PMID: 26325537]
- Burke L et al. Plasma exchange in a real-world ALF cohort. J Hepatol 2025;82:615-21 [PMID: 39362282]
- Wong F et al. CONFIRM. NEJM 2021 [PMID: 33657294]
- Artru F et al. Transplant in ACLF grade 3. J Hepatol 2017 [PMID: 28645736]
- Mathurin P et al. Early transplant for alcoholic hepatitis. NEJM 2011 [PMID: 22070476]
- China L et al. ATTIRE. NEJM 2021 [PMID: 33657293]
- Liver transplantation for ACLF. Lancet Gastroenterol Hepatol 2024 [PMID: 38309288]
- AASLD/AST candidate evaluation guideline. Hepatology 2026 [PMID: 41405234]
- Kyoto consensus on ACLF. Hepatol Int 2025 [PMID: 39961976]
- TERLIVAZ (terlipressin) prescribing information, revised 6/2026